Optimization for Transportation Planning and Periodic Task Scheduling

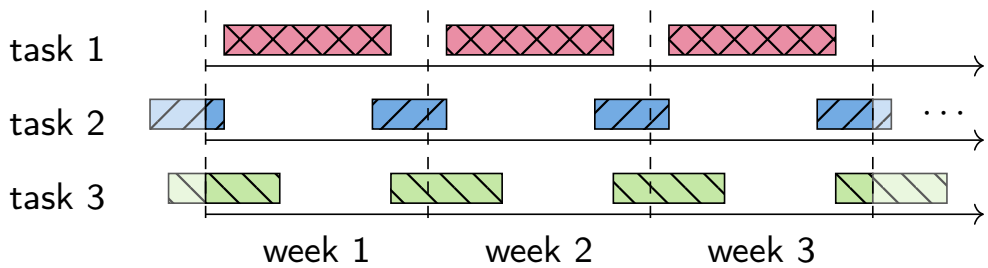
Rail Freight Crew Management under Train Delays, Line Planning, and Balanced Assignments

Supervised by Frédéric Meunier (CERMICS) and Juliette Pouzet (SNCF)

Héloïse Gachet

CERMICS, ENPC, IP Paris | November 3, 2026

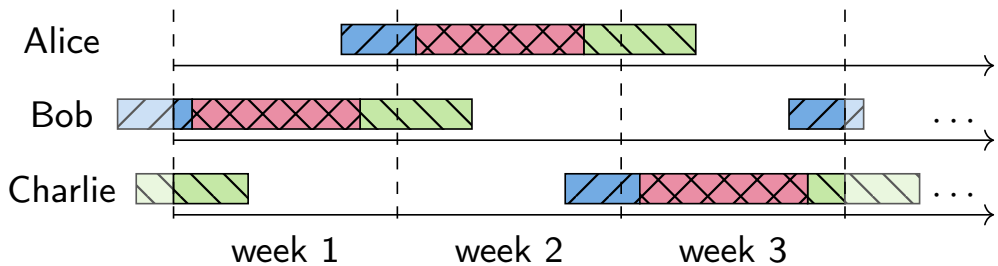
Periodic tasks and assignments



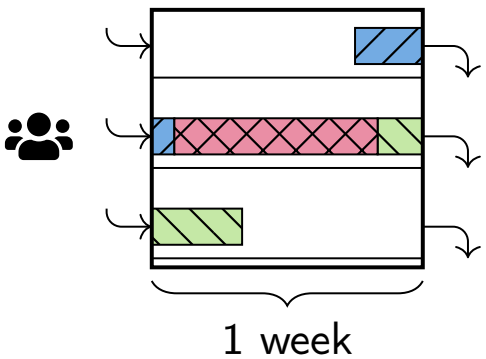
Resources:

Alice
Bob
Charlie
?
?

Aim: Assign tasks to resources for every week



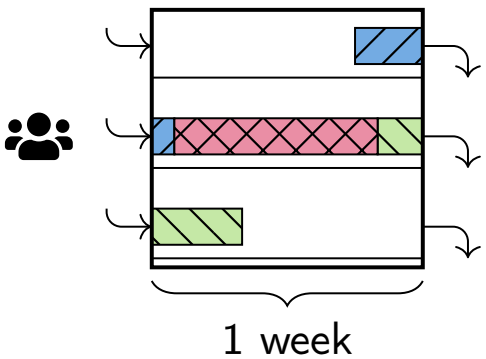
Crew management and balanced assignments



Cyclic rosters:

- ensure balancedness between workers;
- are easy to read and memorize;
- keep workers' proficiency level on every task.

Crew management and balanced assignments

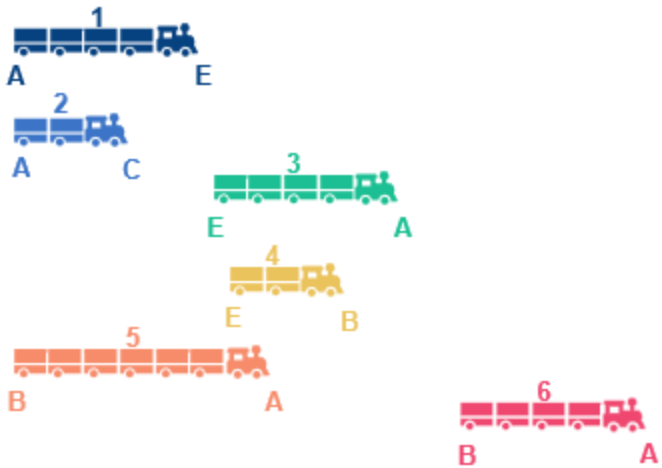


Cyclic rosters:

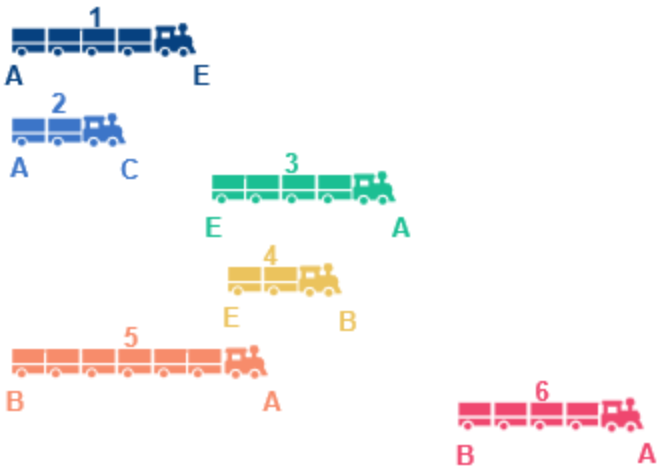
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- are easy to read and memorize;
- keep workers' proficiency level on every task.

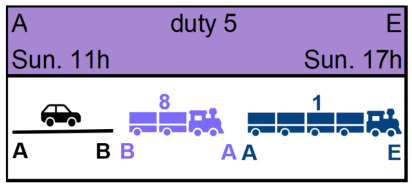
	Crew management	Balanced assignments
Tasks	Trains	Intervals
Structure	Cyclic	Balanced
Number of workers	Not fixed	Fixed
Constraints	Rich	Simple
Goal	Method for efficient solutions	Theorems
Evaluation	Numerical experiments	Proofs

Trains, duties, blocks, and rosters



Trains, duties, blocks, and rosters





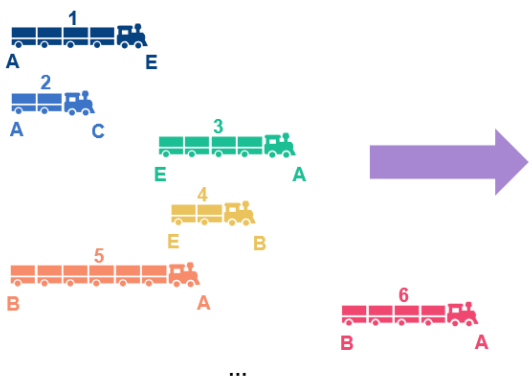
Mon.	Tue.	Wed.
block 1		
duty 7	duty 1	on-call
10h-17h A - A	12h-21h A - A	

Mon.	Tue.	Wed.	Thu.	Fri.	Sat.	Sun.
block 1			R	R	R	block 2
duty 7	duty 1	on-call				duty 5
10h-17h A - A	12h-21h A - A					11h-17h A - E
block 2					R	R
duty 6	duty 2	duty 3	duty 4	on-call		
5h-16h E - A	12h-19h A - B	5h-11h B - A	5h-12h A - A			
R	block 3			R	R	R
	duty 9	duty 10	on-call			
	11h-17h A - E	5h-16h E - A				

Crew management: problem statement

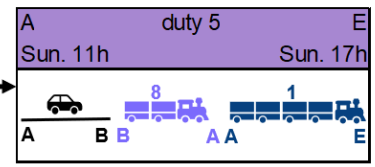
Input. Trains to be driven every week, teams

Output. Covering of trains by rosters with minimum number of working days, one roster per team



Team 1:

Mon.	Tue.	Wed.	Thu.	Fri.	Sat.	Sun.
block 1			R	R	R	block 2
duty 7	duty 1	on-call				duty 5
10h-17h	12h-21h					11h-17h
A - A	A - A					A - E
block 2					R	R
duty 6	duty 2	duty 3	duty 4	on-call		
5h-16h	12h-19h	5h-11h	5h-12h			
E - A	A - B	B - A	A - A			
R	block 3			R	R	R
	duty 9	duty 10	on-call			
	11h-17h	5h-16h				
	A - E	E - A				



Team 2:

Mon.	Tue.	Wed.	Thu.	Fri.	Sat.	Sun.
block 3				R	R	R
duty 11	duty 12	duty 13	on-call			
12h-20h	11h-17h	5h-12h				
B - B	B - B	E - B				

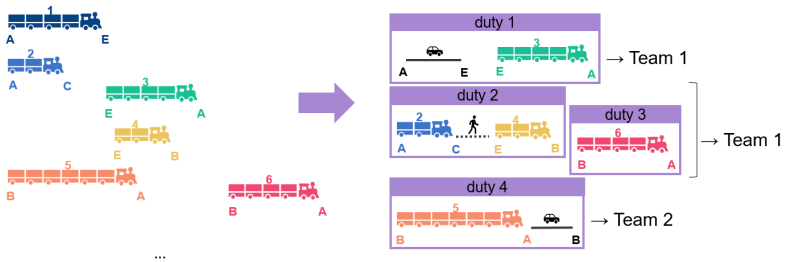
Number of working days = 13

Current heuristic at SNCF: two-stage approach

1. Crew Scheduling

Input. Trains to be driven every week, teams

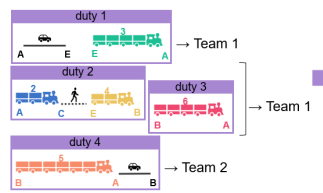
Output. Covering of trains by daily duties with minimum surrogate cost, each duty assigned to a team



2. Crew Rostering

Input. Duties, each assigned to a team

Output. Covering of trains by rosters with minimum number of working days, one roster per team



Team 1:

Mon.	Tue.	Wed.	Thu.	Fri.	Sat.	Sun.
block 1			R	R	block 2	
duty 7	duty 2	duty 3			on-call	duty 5
10h-17h A - A	12h-19h A - B	11h-18h B - A				5h-11h A - E
block 2						
duty 6	duty 1	on-call	duty 4	R	R	R
11h-18h E - A	12h-21h A - A		5h-12h A - A			
block 3						
on-call	duty 9	duty 10	on-call	R	R	R
	11h-17h A - E	5h-16h E - A				

Team 2:

Mon.	Tue.	Wed.	Thu.	Fri.	Sat.	Sun.
block 3						
duty 10	duty 8	duty 9	on-call	R	R	R
12h-20h B - B	11h-17h B - B	5h-12h E - B				

Number of working days = 14

→ Leads to sub-optimal solutions

→ Hard to account for uncertainties

From blocks to rosters

Team 1:

Mon.	Tue.	Wed.
block 1		
duty 7	duty 1	on-call
10h-17h A - A	12h-21h A - A	

Sun.	Mon.	Tue.	Wed.	Thu.	Fri.
block 2					
duty 5	duty 6	duty 2	duty 3	duty 4	on-call
11h-17h A - E	5h-16h E - A	12h-19h A - B	5h-11h B - A	5h-12h A - A	

Tue.	Wed.	Thu.
block 3		
duty 9	duty 10	on-call
11h-17h A - E	5h-16h E - A	

Number of working days = 10

Lemma 3.4.2 (p.45) always possible

Team 1:

Mon.	Tue.	Wed.	Thu.	Fri.	Sat.	Sun.
block 1						block 2
duty 7	duty 1	on-call	R	R	R	duty 5
10h-17h A - A	12h-21h A - A					11h-17h A - E
block 2						
duty 6	duty 2	duty 3	duty 4	on-call	R	R
5h-16h E - A	12h-19h A - B	5h-11h B - A	5h-12h A - A			
R	block 3					
	duty 9	duty 10	on-call	R	R	R
	11h-17h A - E	5h-16h E - A				

Number of working days = 10

Crew management problem:

Input. Trains to be driven every week, teams

Output. Covering of trains by blocks with minimum number of working days, each block assigned to a team

Modeling the crew management problem

- w_b: number of working days in block b
- I: set of teams
- B_i: set of all feasible blocks for team i
- T: collection of trains

$$\begin{aligned} \text{minimize} \quad & \sum_{i \in I} \sum_{b \in B_i} w_b x_{b,i} \\ \text{s.t.} \quad & \sum_{i \in I} \sum_{b \in B_i : t \in b} x_{b,i} \geqslant 1 \qquad \forall t \in T \\ & x_{b,i} \in \{0,1\} \qquad \forall i \in I, \forall b \in B_i \end{aligned}$$

- $x_{b,i} \in \{0,1\}$: block $b \in B_i$ is selected for team $i \in I$

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- $x_{b,i} \in \{0,1\}$: block $b \in B_i$ is selected for team $i \in I$

$$\sum_{i \in I} |B_i| \approx 10^{25}$$

Column generation

Column generation: standard exact method for LP with a large number of variables

- Model with an inherently large number of variables
- Reformulation of a compact model (sometimes) yielding stronger bounds

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To keep it short: iteratively generate "interesting" variables with proof of optimality (without all 10^{25} variables)

- "interesting" $\rightarrow$ solving optimization problem (pricing subproblem)

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For ILP: column generation-based matheuristic (or exact methods: branch-and-price)

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To keep it short: iteratively generate "interesting" variables with proof of optimality (without all 10^{25} variables)

- "interesting" $\rightarrow$ solving optimization problem (pricing subproblem)

For ILP: column generation-based matheuristic (or exact methods: branch-and-price)

For our problem:

Pricing subproblem = Shortest path with constraints

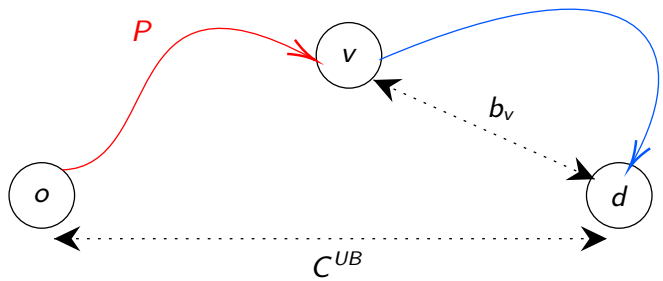
Directed graph:
- vertices = trains
- arc = trains driven successively within a same block
- feasible path = block

Resource Constrained Shortest Path Problem (RCSPP), Section 4.5.2

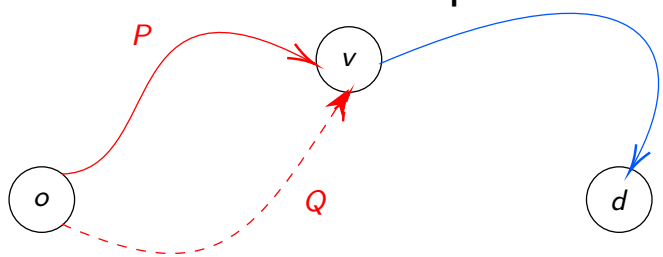
Resource constrained shortest path problem: algorithm

Resolution: path enumeration algorithm using

- Bounds:[1] under-estimate of the resource to reach d



- Dominance:[1] comparison of resources for paths with same last vertex



- Other acceleration techniques: Section 4.5.3

[1] synthetized by Parmentier, A.: **Algorithms for Non-Linear and Stochastic Resource Constrained Shortest Paths**, ***Mathematical Methods of Operations Research***. **(2017)**

Numerical experiments

- Run with:
  - Intel(R) Xeon(R) CPU E5-2690 v2 @ 3.00GHz processor
  - programming language Java 18
  - solver CPLEX 22.11
- Integer linear program time limit 1 h

	Two-stage approach		Integrated approach			
	Time	Best found solution	Time	LB		Best found solution
116 trains	1 min	50	1 h 8 min	40.3	(-18%)	41
280 trains	3 min	228	1 h 9 min	205.3	(-9%)	207
926 trains	26 min	832	6 h 3 min	711.9	(-13%)	724
1,815 trains	64 min	1,633	10 h 2 min	1,395.0	(-10%)	1,477

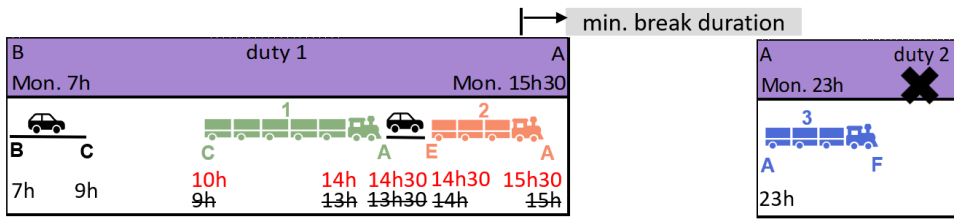
Uncertainties at SNCF: delay propagation and disruptions

Train delays of two kinds:

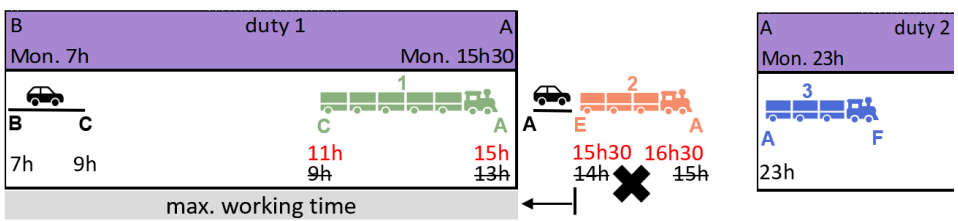
- Primary delays: exogenous (e.g., shunting yard incident, engine failure, etc.)
- Secondary delays: endogenous, from delay propagation within duties

When delays exceed some threshold:

- a break might not meet the minimal duration. In this case, the next duty is disrupted;



- the working time within the duty might be too long. In this case, one or multiple trains are disrupted.



Uncertainties at SNCF: fragility score

Assumption:

- primary delays distributions are known.

H: random variable for arrival times (with primary and secondary delays)
$\sigma(h, b)$: number of disruptions in the realization $H = h$

Block "fragility" score:

$$\text{fragility}(b) := \mathbb{E}\Big[\sigma(H, b)\Big]$$

Roster "fragility" score:

$$\text{fragility}(r) := \sum_{b \in r} \text{fragility}(b)$$

Stochastic crew management: problem statement and model

Stochastic crew management problem:
Input. Trains to be driven every week, teams, threshold S
Output. Covering of trains by blocks with minimum number of working days, each block assigned to a team, such that the overall fragility is below the threshold S

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$$\begin{aligned}
\text{minimize} \quad & \sum_{i \in I} \sum_{b \in B_i} w_b x_{b,i} \\
\text{s.t.} \quad & \sum_{i \in I} \sum_{b \in B_i : t \in b} x_{b,i} \geqslant 1 && \forall t \in T \\
& \sum_{i \in I} \sum_{b \in B_i} \mathbb{E}[\sigma(H,b)] x_{b,i} \leqslant S \\
& x_{b,i} \in \{0,1\} && \forall i \in I, \forall b \in B_i
\end{aligned}$$

- $x_{b,i} \in \{0,1\}$: block $b \in B_i$ is selected for team $i \in I$

Column generation and RCSPP

Assumptions:

- primary delays are independent variables (not necessarily identically distributed).
- their supports are finite.

Marginal modifications of the column generation-based algorithm:

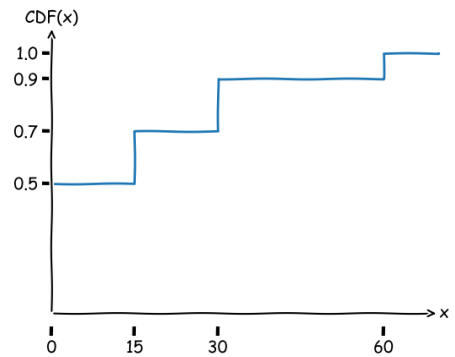
- Farkas lemma based routine to tackle infeasiblilities
- RCSPP bounds based on the usual stochastic order $\leqslant_{\mathrm{st}}$
- Handling delay propagation along a path via distributions convolutions

Numerical experiments: stochastic case

- Run with same setup as the deterministic case
- Column generation time limit 24 h: reached for every instance (stochastic case)
- Integer linear program time limit 1 h

- Threshold S: 50% of two-stage solution

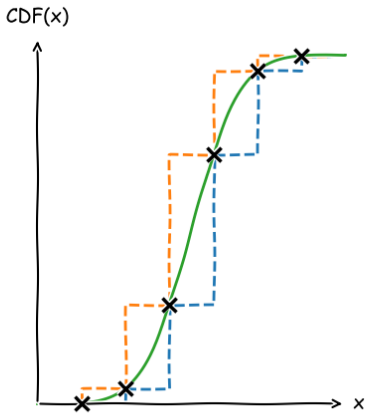
Primary delay cumulative distribution function:



	Two-stage approach		Integrated approach: deterministic case	
	Best found solution	Fragility	Best found solution	Fragility
116 trains	50	0.970	41	2.748
280 trains	228	2.450	207	6.440
926 trains	832	8.538	724	28.460
1,815 trains	1,633	14.900	1,477	58.980

	Integrated approach: stochastic case			
	Best found solution			Fragility
116 trains	(-12%)	44	(-55%)	0.437
280 trains	(-8%)	210	(-51%)	1.202
926 trains	(-11%)	741	(-52%)	4.081
1,815 trains	(-10%)	1,477	(-50%)	7.432

From finite support delay distributions to any distribution

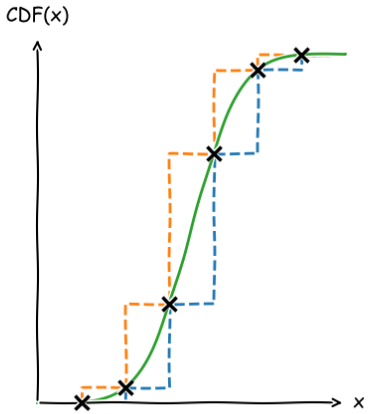


$P'_t \leqslant_{\text{st}} P_t \leqslant_{\text{st}} P''_t$ for every $t \in T$

optimal value under $(P'_t)_t \leqslant$ optimal value under $(P_t)_t$

optimal value under $(P_t)_t \leqslant$ optimal value under $(P''_t)_t$

From finite support delay distributions to any distribution



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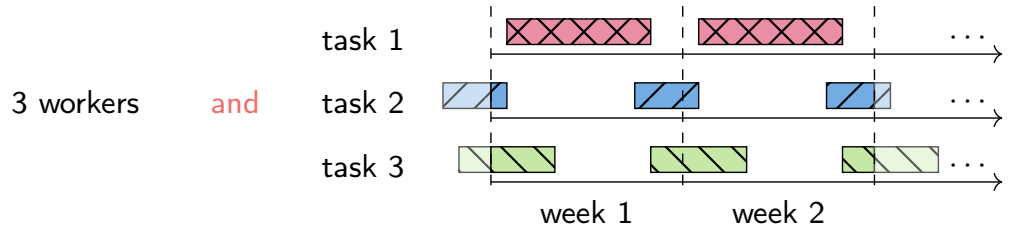
- Original primary delay distributions $(\bar{P}_t)_t$
- Sequence of alternative primary delay distributions $(P_{t,n})_{t,n}$ such that
  - $P_{t,n} \leqslant_{\text{st}} \bar{P}_t$ for every t and n
  - $P_{t,n} \xrightarrow{\text{d}} \bar{P}_t$ for every t

Theorem 5.6.2, p.75

Suppose that the problem is feasible under $(\bar{P}_t)_t$. Then there exists N such that, for all $n \geqslant N$, the sets of feasible solutions under $(P_{t,n})_t$ and under $(\bar{P}_t)_t$ coincide (and the same holds for the set of optimal solutions).

Motivation

- Tasks $[n]$ to be performed at the same time every week
- Number q of indistinguishable workers

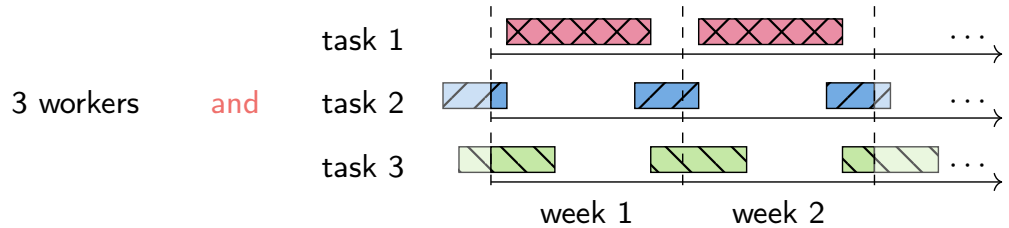


Aim. Find a feasible assignment of tasks to workers so that each worker performs each task with same asymptotic frequency

→ Balanced assignment

Motivation

- Tasks $[n]$ to be performed at the same time every week
- Number q of indistinguishable workers



Aim. Find a feasible assignment of tasks to workers so that each worker performs each task with same asymptotic frequency
→ Balanced assignment

Assignment: $f : [n] \times \mathbb{Z}_{>0} \to [q]$, where $f(i, r) = j$ means worker j performs task i on week r

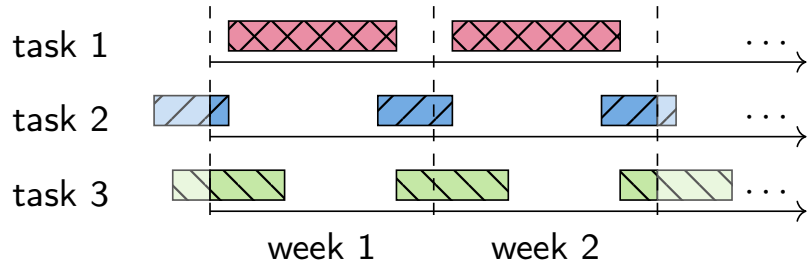
Feasible assignment: f is non overlapping.

Balanced assignment:

$$f \text{ feasible} \quad \text{and} \quad \lim_{t \to +\infty} \frac{1}{t} \left| \{ r \in [t] : f(i, r) = j \} \right| = \frac{1}{q} \qquad \forall i, j$$

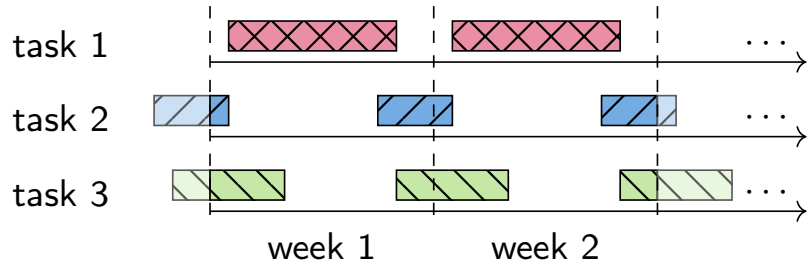
Feasible and balanced assignments: examples

3 workers and

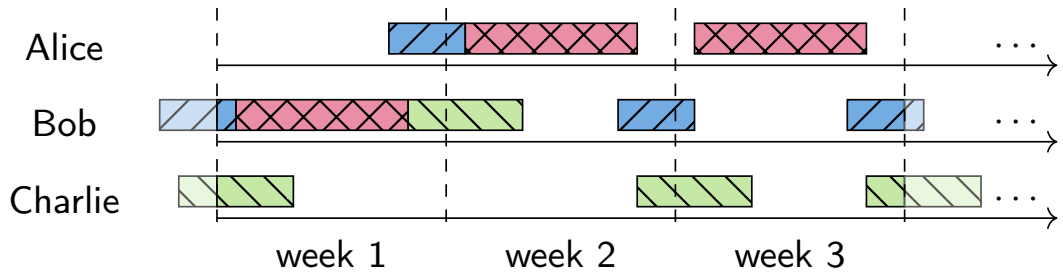


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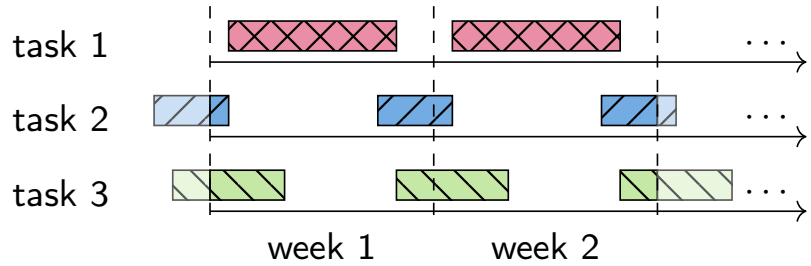


Feasible assignment:

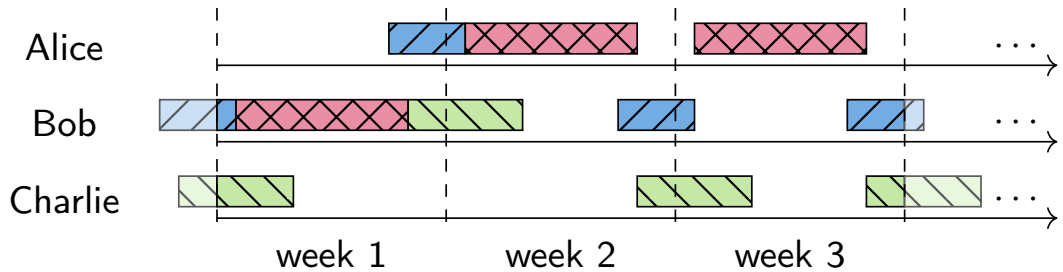


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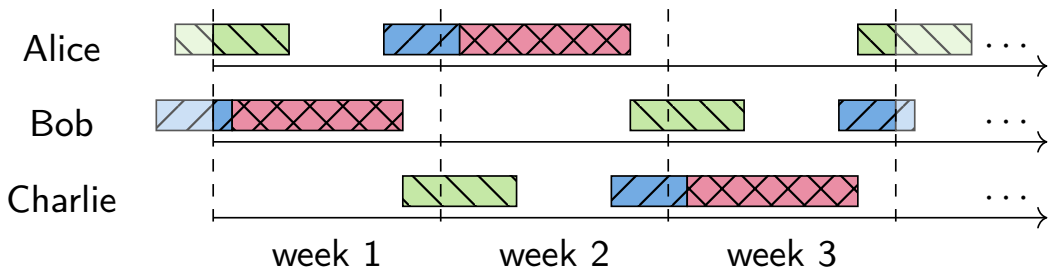
3 workers and



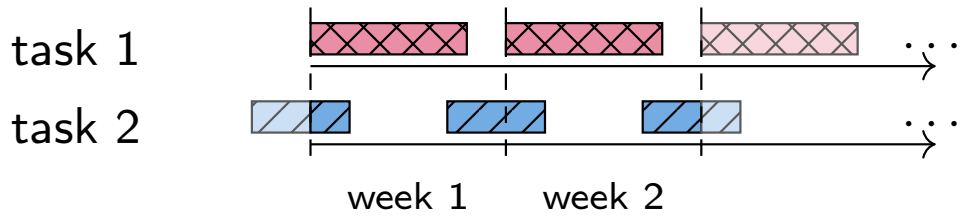
Feasible assignment:



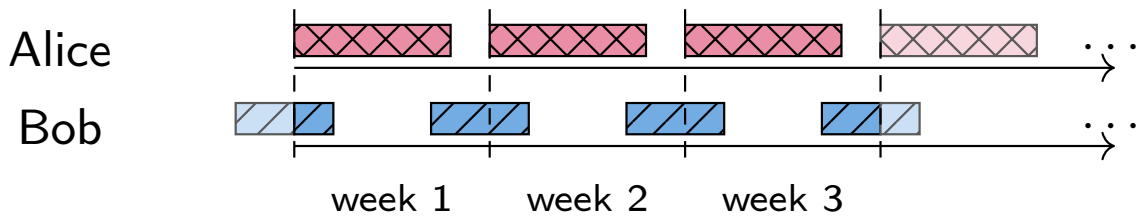
Balanced assignment:



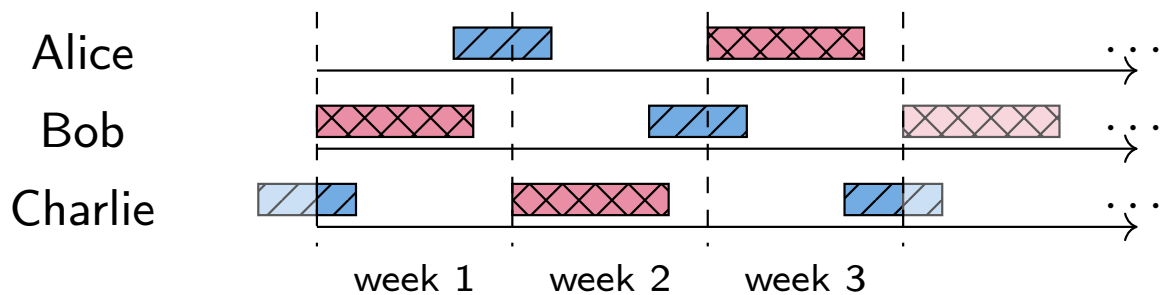
Balanced assignment: unachievable case



With 2 workers

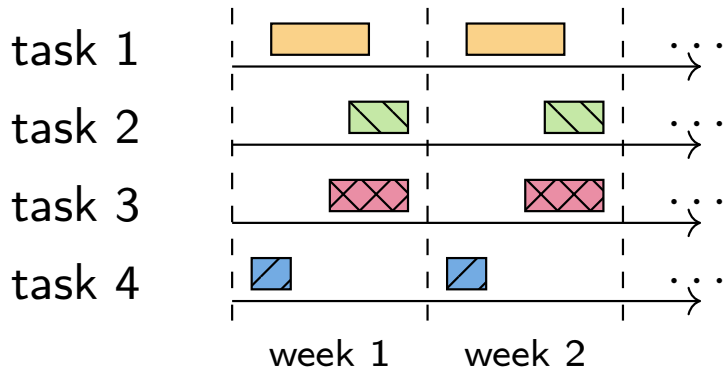


With 3 workers

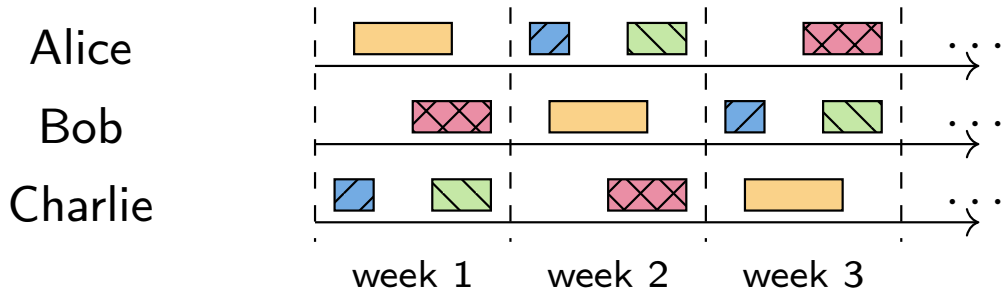


Balanced assignment: trivial case

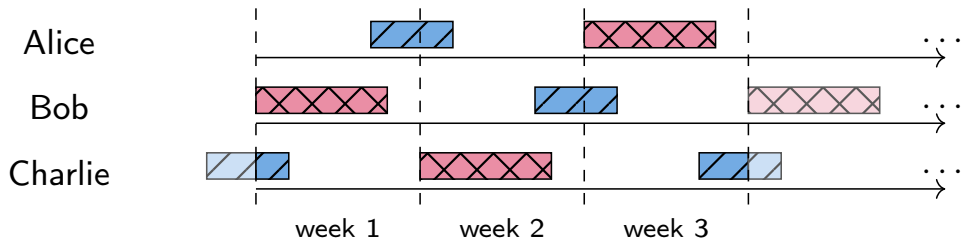
3 workers and



Balanced assignment:



Main results



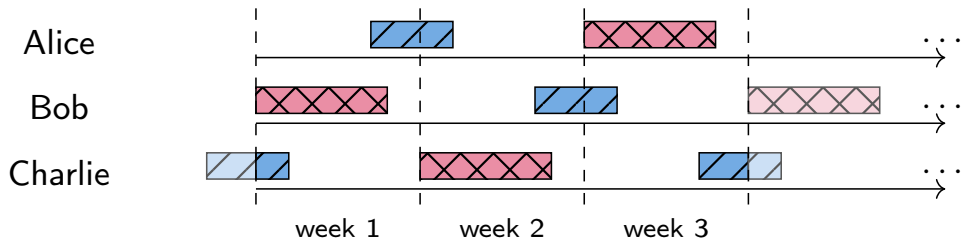
$$\lim_{t \to +\infty} \frac{1}{t} \left| \{ r \in [t] : f(i,r) = j \} \right| = \frac{1}{q} \qquad \forall i, j$$

There exists a balanced assignment.

$\Downarrow$

There exists a feasible assignment with a worker performing each task at least once.

Main results



$$\lim_{t \to +\infty} \frac{1}{t} \left| \{ r \in [t] : f(i,r) = j \} \right| = \frac{1}{q} \qquad \forall i, j$$

There exists a balanced assignment.

$\Updownarrow$

There exists a feasible assignment with a worker performing each task at least once.

Main results

Theorem 7.1.1, p.106

The following statements are equivalent:

1. There exists a feasible assignment with a worker performing each task at least once.
2. There exists a balanced assignment.
3. There exists a periodic balanced assignment with period q.

q: number of workers

Algorithmic proof with polynomiality results:

- Deciding existence of a balanced assignment
- Computing a periodic balanced assignment with period q

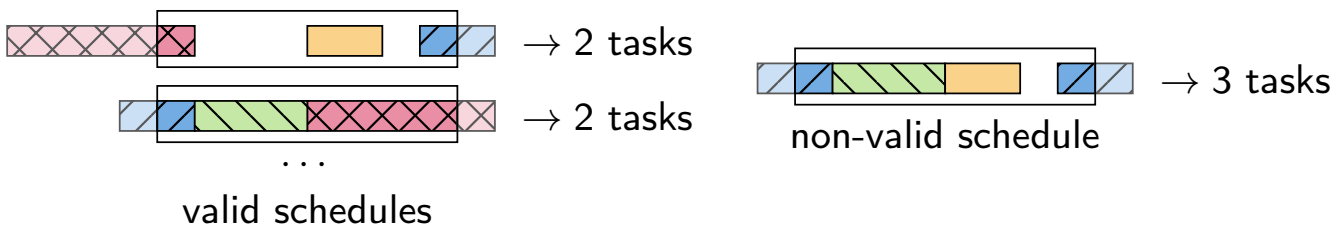
Extended version: setting

Input. n tasks to be repeated every week, q workers, set $\mathcal{S}$ of valid weekly schedules

Feasible assignment: f non overlapping and induces schedules in $\mathcal{S}$

Examples:

- workers perform at most M tasks per week.
  with $M = 2$



- workers work at most H hours per week.
- when q tasks have to be performed at time zero, workers cannot teleport.
- ...

Balanced assignments: results

Theorem 8.1.2, p.126

If there exists a feasible assignment with a worker performing each task at least once and q tasks have to be performed at time zero, then there exists a balanced assignment.

q: number of workers

Examples where the additionnal condition is needed (Fig. 8.1, p.125)

Theorem 8.1.1, p.124

If there exists a balanced assignment, then there exists a periodic balanced assignment with period bounded by $2(q-1)q!$.

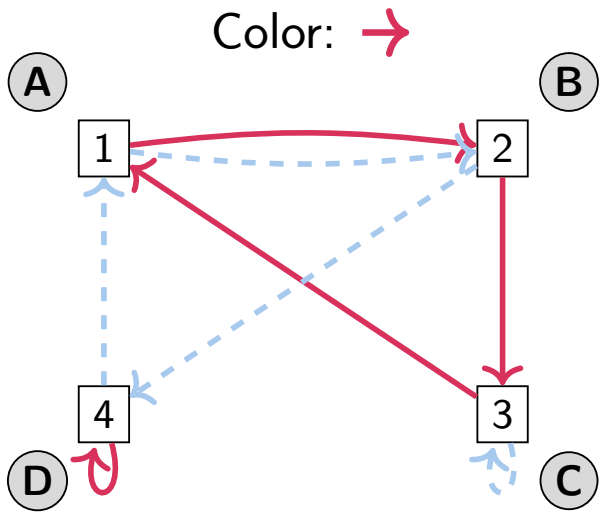
- Existence of a balanced assignment decidable

A problem of pebbles on a graph

Consider:

- graph $D = (V, A)$ with colored arcs, each color forming vertex disjoint cycles covering V
- one pebble on each vertex

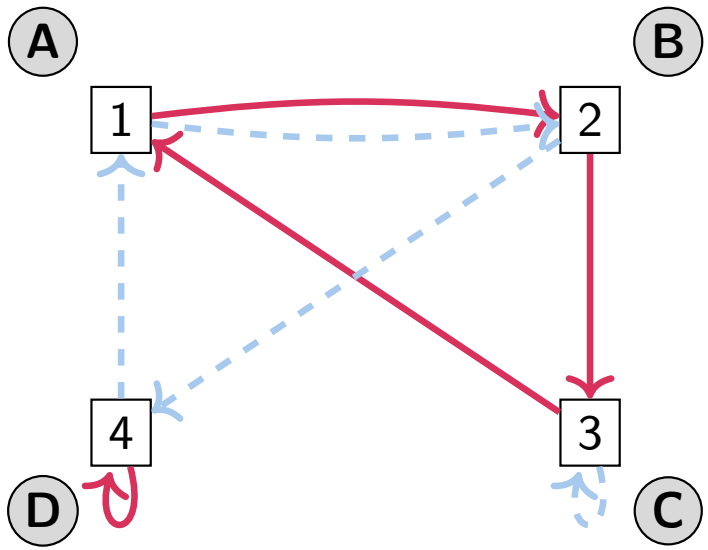
Sequence of colors defines sequence of moves of the pebbles



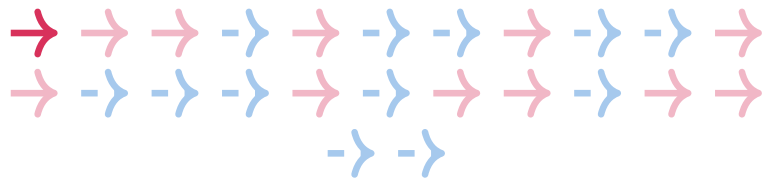
- Vertex = task at time zero
- Arc = valid schedule
- Pebble = worker
- Color c = partition of the tasks into schedules

Balanced sequence of colors: example

Aim. Find a sequence of colors making each pebble visit each arc with same asymptotic frequency.



Sequence of colors:



Arcs visiting:

	1→2	2→3	3→1	4→4	1⇢2	2⇢4	3⇢3	4⇢1
A	**1**	0	0	0	0	0	0	0
B	0	**1**	0	0	0	0	0	0
C	0	0	**1**	0	0	0	0	0
D	0	0	0	**1**	0	0	0	0

Balanced sequence of colors: result

Proposition 8.3.1, p.129

If D is connected, then there exists a sequence of colors making each pebble visit each arc with same asymptotic frequency.

Two proofs:

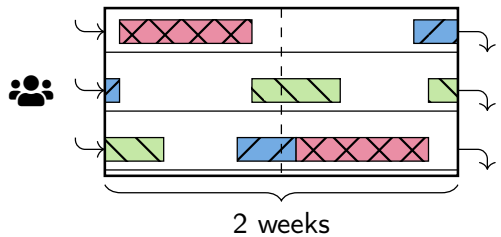
- Markov chains: non constructive
- Eulerian cycle in the Cayley graph associated to the set Σ of colors (seen as permutations of V):
  - Contructive proof
  - Additionnal result: the sequence of colors can be made periodic (period upper bounded by $|A| \times |\langle\Sigma\rangle|/|V|$)

Balanced assignments: graph connected with $2(q-1)$ colors

$\rightarrow$ period upper bounded by $2(q-1)q!$

q: number of workers

What about cyclic rosters?



Corollary 8.5.5, p.135

Suppose q is a prime and there exists a balanced assignment. There exists then a cyclic roster on a k-week basis, with $k \leqslant q! - 1$.

q: number of workers

Using our graph framework along with:

- Group theory tools: Cauchy theorem, orbit-stabilizer theorem

Overall thesis contributions

- Integrating crew scheduling and crew rostering with train delays for rail freight
  - (CHAPTER 4) Integrated crew management
  - (CHAPTER 5) Integrated stochastic crew management
- Balanced assignments of periodic tasks
  - (CHAPTER 7) Basic version: non-overlapping balanced assignments
  - (CHAPTER 8) Extended version: balanced assignments with valid weekly schedules

Overall thesis contributions

- Integrating crew scheduling and crew rostering with train delays for rail freight
  - (CHAPTER 3) Crew scheduling and crew rostering at SNCF
  - (CHAPTER 4) Integrated crew management
  - (CHAPTER 5) Integrated stochastic crew management
- Branch-and-price(-and-cut) for line planning with minimal traveling time
  - (CHAPTER 6) Branch-and-price
- Balanced assignments of periodic tasks
  - (CHAPTER 7) Basic version: non-overlapping balanced assignments
  - (CHAPTER 8) Extended version: balanced assignments with valid weekly schedules
  - (CHAPTER 9) Complementary discussions

Perspectives and open questions

Crew management	Balanced assignments
Multi-objective optimization for the trade-off between fragility and number of working days	Asymptotic bounds from group theory for cyclic rosters
Can we establish an absolute bound on the rank N from which optimal solutions coincide for original and approximate distribution functions?	Is the bound $2(q-1)q!$ for periodic balanced assignments tight for a single instance?

Thank you!

Duty constraints

R1	a task ending at a location must be followed by a task starting at the same location
R2	a team can only drive trains compatible to its set of skills
R3	maximal working range in a daytime duty with a short break = 11h
R4	maximal working range in a daytime duty without a short break = 8h
R5	maximal working range in a nighttime duty = 8h
R6	for a nighttime duty with least 5h of work and 2h of nighttime work, maximal effective work duration = 9h
R7	short breaks cannot happen at nighttime
R8	"meal" short breaks must happen in a dedicated interval, be it at lunch or at dinner
R9	minimal work duration in a duty before and after a short break = 1h30
R10	minimal short break duration = 1h30
R11	given start-of-duty duration (depending on the first task)
R12	given end-of-duty duration (depending on the last task)

Block constraints

R13	back to team's residency at most every two duties
R14	a duty ending at a location must be followed by a duty starting at the same location
R15	a team can only drive trains compatible to its set of skills
R16	maximal number of days = 6
R17	maximal number of on-calls and duties = 6
R18	maximal number of night duties = 6
R19	a duty starting a block before 6:00 must be preceded by a short on-call
R20	a duty ending a block after 19:00 must be followed by a short on-call
R21	at the team's residency, between two duties with a long break smaller than 38h, there must be a short on-call
R22	minimal long break duration at team's residency = 14h
R23	minimal long break not at team's residency = 8h
R24	a long break shorter than 9h is followed by a long break longer than 15h

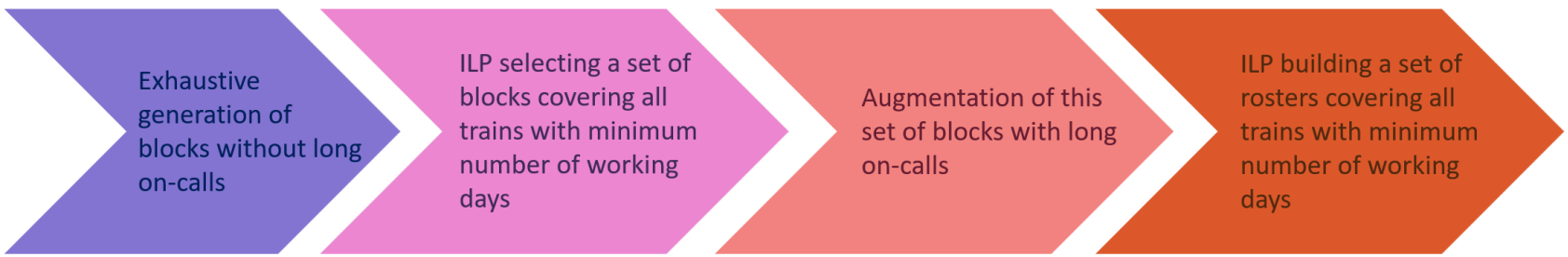
Roster constraints

R25	minimal number of annual rest days = 120
R26	minimal number of annual consecutive rest days = 52
R27	minimal number of annual Sundays that are rest days = 22
R28	minimal number of annual consecutive rest days with a Sunday = 14
R29	minimal number of annual weekends where both days are rest days = 12
R30	maximal number of consecutive worked Sundays = 3
R31	minimal number of rest days between two blocks = 1 if it is a Sunday, 2 otherwise
R32	maximal number of rest days between two blocks = 3
R33	minimal number of rest days after a block with 6 working days = 2
R34	minimal/maximal number of days in each block of long on-calls = 3/5
R35	between two successive blocks with the first block ending with a duty and the second starting with a duty, there must be at least 14h + 24h $\times$ number of full days between the two blocks

Crew scheduling algorithm at SNCF



Crew rostering algorithm at SNCF



Current ILP at SNCF

$$
\begin{aligned}
\text{minimize} \quad & \sum_{b \in \widetilde{B}_i} w_b x_b \\
\text{s.t.} \quad & \sum_{a \in A : a = (b, b')} y_a && = x_b \forall b \in \widetilde{B}_i \\
& \sum_{a \in A : a = (b', b)} y_a && = x_b \forall b \in \widetilde{B}_i \\
& \sum_{a \in A} |\text{days}(a)| y_a && \geqslant \frac{116}{365} \left(\sum_{a \in A} |\text{days}(a)| y_a + \sum_{b \in \widetilde{B}_i} |\text{days}(b)| x_b \right) \\
& \sum_{\substack{a \in A \\ |\text{days}(a)| \geqslant 2}} y_a && \geqslant \frac{52}{365} \left(\sum_{a \in A} |\text{days}(a)| y_a + \sum_{b \in \widetilde{B}_i} |\text{days}(b)| x_b \right) \\
& \sum_{\substack{a \in A \\ \text{SUN} \in \text{days}(a)}} y_a && \geqslant \frac{22}{365} \left(\sum_{a \in A} |\text{days}(a)| y_a + \sum_{b \in \widetilde{B}_i} |\text{days}(b)| x_b \right) \\
& \sum_{\substack{a \in A \\ \text{SUN} \in \text{days}(a) \\ |\text{days}(a)| \geqslant 2}} y_a && \geqslant \frac{14}{365} \left(\sum_{a \in A} |\text{days}(a)| y_a + \sum_{b \in \widetilde{B}_i} |\text{days}(b)| x_b \right) \\
& \sum_{\substack{a \in A \\ \{\text{SAT}, \text{SUN}\} \subseteq \text{days}(a)}} y_a && \geqslant \frac{12}{365} \left(\sum_{a \in A} |\text{days}(a)| y_a + \sum_{b \in \widetilde{B}_i} |\text{days}(b)| x_b \right) \\
& \sum_{b \in \widetilde{B}_i : t \in b} x_b && \geqslant 1 \forall t \in \widetilde{T}_i \\
& x_b \in \{0, 1\} && \forall b \in \widetilde{B}_i \\
& y_a \in \{0, 1\} && \forall a \in A
\end{aligned}
$$

Alternative ILP—Proposition 3.4.1

$$\sum_{\substack{b' \in \widetilde{B}_i \\ d \leqslant d_{b'} \leqslant f}} z_{b',r,i} + 2 \sum_{\substack{b \in \widetilde{B}_i \\ d_b = d, f_b = f}} z_{b,r,i} \leqslant 3 \qquad \forall i \in I, r \in [N], (d,f) \in [7]^2 : d \leqslant f$$

$$\sum_{\substack{b' \in \widetilde{B}_i \\ d_{b'} \geqslant d}} z_{b',r,i} + 2 \sum_{\substack{b \in \widetilde{B}_i \\ d_b = d, f_b = f}} z_{b,r,i} \leqslant 3 \forall i \in I, r \in [N], (d,f) \in [7]^2 : d \geqslant f$$

$$\sum_{\substack{b' \in \widetilde{B}_i \\ d_{b'} \leqslant f}} z_{b',r+1,i} + 3 \sum_{\substack{b \in \widetilde{B}_i \\ d_b \geqslant f_b, f_b = f}} z_{b,r,i} \leqslant 3 \forall i \in I, r \in [1, N-1], f \in [7]$$

$$\sum_{\substack{b' \in \widetilde{B}_i \\ d_{b'} \leqslant f}} z_{b',1,i} + 3y_{r,i} + 3 \sum_{\substack{b \in \widetilde{B}_i \\ d_b \geqslant f_b, f_b = f}} z_{b,r,i} \leqslant 6 \forall i \in I, r \in [N], f \in [7]$$

Alternative ILP—Proposition 3.4.1

$$\sum_{r=j}^{j+N_{SUN}^{max}} \sum_{\substack{b\in B_i \\ SUN\in \mathbf{days}(b)}} z_{b,r,i} \leqslant N_{SUN}^{max} \qquad \forall i \in I, j \in [1, N - N_{SUN}^{max}]$$

$$\sum_{r\in[j-N_{SUN}^{max}+k,j]\cup[k]} \sum_{\substack{b\in B_i \\ SUN\in \mathbf{days}(b)}} z_{b,r,i} \leqslant N_{SUN}^{max} + 1 - y_{j,i} \qquad \forall i \in I, j \in [N], k \in [N_{SUN}^{max}]$$

$$(1 - \frac{116}{365}) \sum_{r\in[N]} x_{r,i} \geqslant \sum_{r\in[N]} \sum_{b\in B_i} |\mathrm{days}(b)| z_{b,r,i} \qquad \forall i \in I$$

$$\sum_{r\in[N]} \sum_{b\in B_i} z_{b,r,i} \geqslant \frac{52}{365} \sum_{r\in[N]} 7x_{r,i} + \sum_{r\in[N]} w_{r,i} \qquad \forall i \in I$$

$$(1 - \frac{7*22}{365}) \sum_{r\in[N]} x_{r,i} \geqslant \sum_{r\in[N]} \sum_{\substack{b\in B_i \\ SUN\in \mathbf{days}(b)}} z_{b,r,i} \qquad \forall i \in I$$

$$(1 - \frac{7*14}{365}) \sum_{r\in[N]} x_{r,i} \geqslant \sum_{r\in[N]} w_{r,i} + \sum_{r\in[N]} \sum_{\substack{b\in B_i \\ SUN\in \mathbf{days}(b)}} z_{b,r,i} \qquad \forall i \in I$$

$$(1 - \frac{7*12}{365}) \sum_{r\in[N]} x_{r,i} \geqslant \sum_{r\in[N]} \sum_{\substack{b\in B_i \\ SAT \text{ or } SUN\in \mathbf{days}(b)}} z_{b,r,i} \qquad \forall i \in I$$

Alternative ILP—Proposition 3.4.1

$$2x_{r,i} \geqslant \sum_{b \in B_i} z_{b,r,i} \qquad \forall i \in I, r \in [N]$$

$$3x_{r,i} \geqslant \frac{1}{2} \sum_{b \in B_i} z_{b,r+1,i} + \sum_{\substack{b \in B_i \\ d_b > f_b \\ f_b = f}} z_{b,r-1,i} + \frac{1}{2} \sum_{\substack{b \in B_i \\ d_b \leqslant f+2}} z_{b,1,i} - 1 \qquad \forall i \in I, r \in [N], f \in [7]$$

$$x_{r,i} \geqslant x_{r+1,i} \qquad \forall i \in I, r \in [N]$$

$$y_{r,i} = x_{r,i} - x_{r+1,i} \qquad \forall i \in I, r \in [N]$$

$$w_{r,i} \geqslant \sum_{\substack{b \in B_i \\ d_b \leqslant f_b \\ f_b = 6}} z_{b,r,i} + \sum_{\substack{b \in B_i \\ d_b = 1}} z_{b,(r \mod N)+1,i} - 1 \qquad \forall i \in I, r \in [N]$$

$$w_{r,i} \geqslant \sum_{\substack{b \in B_i \\ d_b \leqslant f_b \\ f_b = 6}} z_{b,r,i} + \sum_{\substack{b \in B_i \\ d_b = 1}} z_{b,1,i} + 2y_{r,i} - 2 \qquad \forall i \in I, r \in [N]$$

Alternative ILP—Proposition 3.4.1

$$\sum_{\substack{b\in B_i\\ d_b\leqslant f_b\\ f_b=f}} z_{b,r,i} + \sum_{\substack{b\in B_i\\ d_b>f_b\\ f_b=f}} z_{b,r-1,i} \leqslant 1 - \sum_{d=1}^{1+\mathbb{1}_{(f\neq 6)}} \left(\sum_{\substack{b'\in B_i\\ d_{b'}=d'\\ d':=(f+d-1 \mod 7)+1\\ r':=((r+(f+d-1)//7)-1 \mod N)+1}} z_{b',r',i} \right) \qquad \forall i\in I, r\in[2,N], f\in[7]$$

$$\sum_{\substack{b\in B_i\\ d_b\leqslant f_b\\ f_b=f}} z_{b,r,i} + \sum_{\substack{b\in B_i\\ d_b>f_b\\ f_b=f}} z_{b,r-1,i} \leqslant \sum_{d=1}^{3} \left(\sum_{\substack{b'\in B_i\\ d_{b'}=d'\\ d':=(f+d \mod 7)+1\\ r':=((r+(f+d)//7)-1 \mod N)+1}} z_{b',r',i} \right) - \frac{1}{N} \left(\sum_{\substack{b'\in B_i\\ d_{b'}\geqslant f+3}} z_{b',r,i} + \sum_{k\geqslant r+1} z_{b,k,i} \right) + 1 \qquad \forall i\in I, r\in[2,N-1], f\in[7]$$

$$\sum_{\substack{b\in B_i\\ d_b\leqslant f_b\\ f_b=f}} z_{b,r,i} + \sum_{\substack{b\in B_i\\ d_b>f_b\\ f_b=f}} z_{b,r-1,i} + 2y_{r,i} - 2 \leqslant \sum_{d=1+\mathbb{1}_{(f\neq 6)}}^{3} \sum_{\substack{b'\in B_i\\ d_{b'}=f+d+1-7}} z_{b',1,i} \qquad \forall i\in I, r\in[2,N], f\in[5,7]$$

$$\sum_{\substack{b\in B_i\\ d_b\leqslant f_b\\ f_b=f}} z_{b,1,i} + \sum_{\substack{b\in B_i\\ d_b>f_b\\ f_b=f}} z_{b,r,i} + 2y_{r,i} - 2 \leqslant \sum_{d=1+\mathbb{1}_{(f\neq 6)}}^{3} \left(\sum_{\substack{b'\in B_i\\ d_{b'}=d'\\ d':=(f+d \mod 7)+1\\ r':=((f+d)//7)+1}} z_{b',r',i} \right) \qquad \forall i\in I, r\in[1,N], f\in[7]$$

Alternative ILP—Proposition 3.4.1

$\Delta_{f,r,d,r'} = (24 \times (d + 7(j-i) - f - 1) + 14) \times 60$
$\mu = (N+1) \times 24 \times 60$

$$\sum_{\substack{b' \in B_i \\ d_{b'}^{\mathbf{duty}} = d}} h_{b'}^{\mathbf{start}} z_{b',(r'-1) \mod N+1,i} - \left(\sum_{\substack{b \in B_i \\ d_b \leqslant f_b \\ f_b^{\mathbf{duty}} = f}} h_b^{\mathbf{end}} z_{b,r,i} + \sum_{\substack{b \in B_i \\ \mathbf{end}(b) \in \mathbf{duty} \\ d_b > f_b \\ f_b^{\mathbf{duty}} = f}} h_b^{\mathbf{end}} z_{b,r-1,i} \right)$$

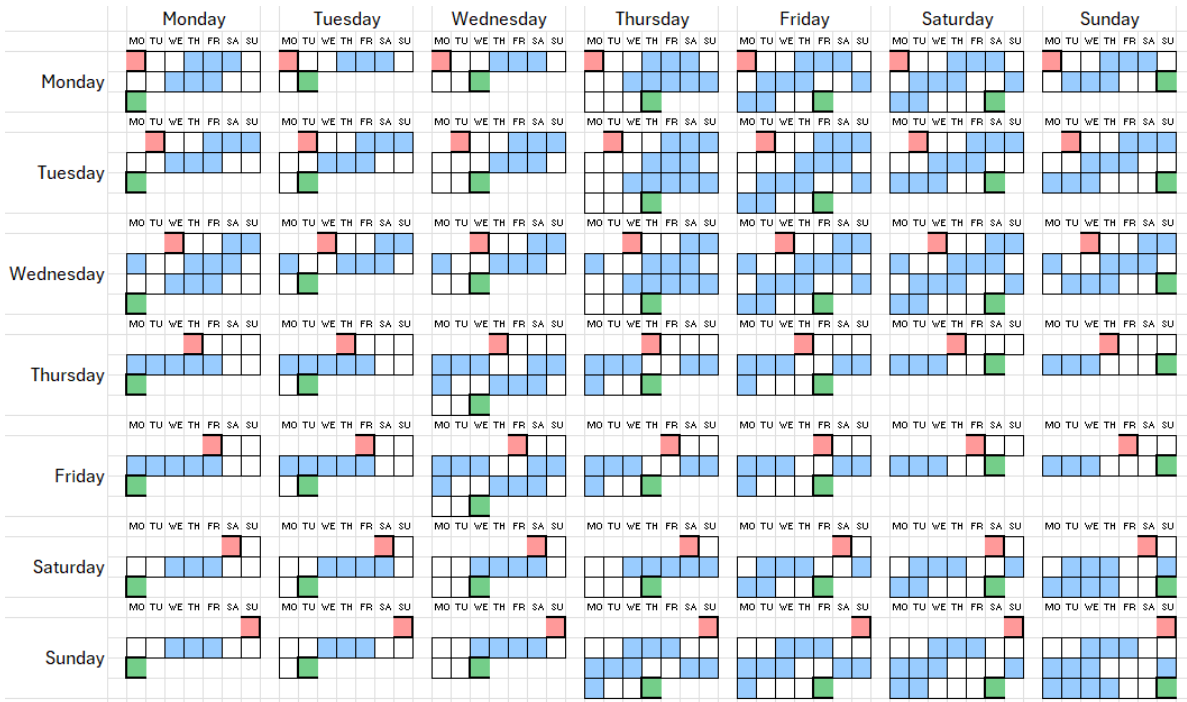
$$\geqslant (\mu + \Delta_{f,r,d,r'}) \sum_{\substack{b' \in B_i \\ d_{b'}^{\mathbf{duty}} = d}} z_{b',(r'-1) \mod N+1} + \Delta_{f,r,d,r'} \left(\sum_{\substack{b \in B_i \\ d_b \leqslant f_b \\ f_b^{\mathbf{duty}} = f}} z_{b,r,i} + \sum_{\substack{b \in B_i \\ d_b > f_b \\ f_b^{\mathbf{duty}} = f}} z_{b,r-1,i} \right) - \mu - \Delta_{f,r,d,r'} - \mu \sum_{k \in [r,r']} \sum_{\substack{b'' \in B_i \setminus G^D \\ b'' \subset [7r+f, 7r'+d]}} z_{b'',k,i}$$

$$\forall i \in I, r \in [N], r' \in [i, i+N], (d,f) \in [7]^2, f \leqslant d + 7(j-i)$$

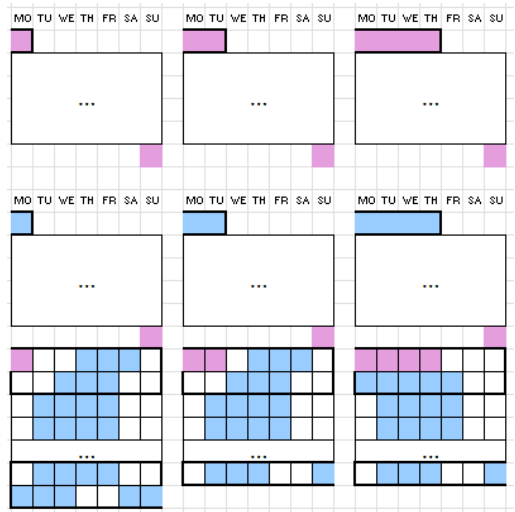
Proof of Lemma 3.4.2

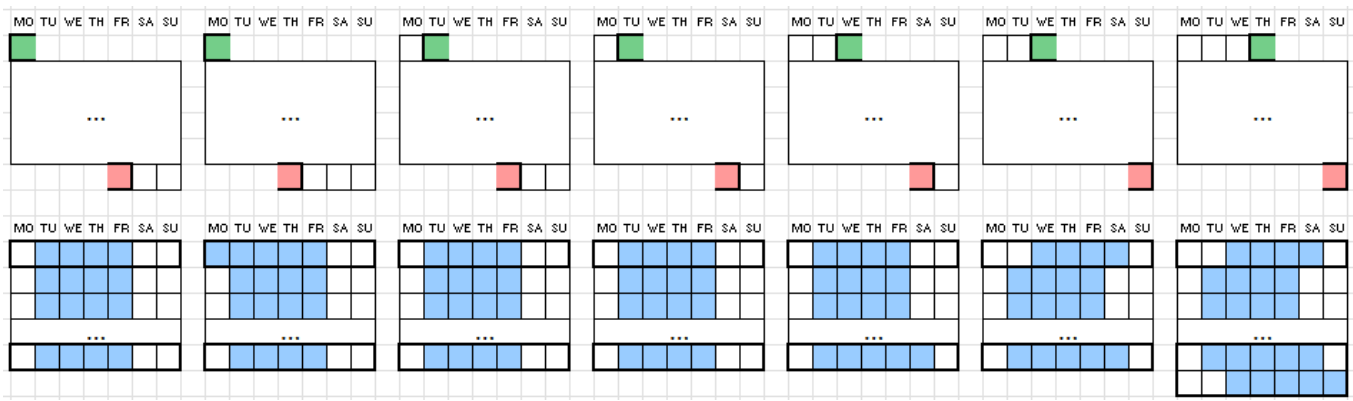
Lemma 3.4.2

Given blocks $b_1, \ldots, b_k$, there exists a roster covering $b_1, \ldots, b_k$, without adding any working days, and with a number of rows upper bounded by $13k$.



Proof of Lemma 3.4.2





Algorithms comparison

Instance	Number working days	Current algorithm		Alternative ILP			Alternative ILP: binary search				
		Time(s)	Rows	Rows UB	Time(s)	Rows	Rows UB	Rows LB	Time(s)	Iterations	Rows
AF	34	**1.0**	8	17	3.3	8	17	6	7.5	4	8
AMB-2	19	0.7	5	10	0.9	5	10	2	**0.5**	3	5
BGW-2	48	**2.5**	11	15	2.9	11	15	8	8.2	3	11
BGW-3	42	12.8	10	12	**2.5**	10	12	5	4.9	3	10
BYE	18	0.1	4	6	0.3	4	6	3	**0.0**	2	4
CHA	7	0.1	3	5	0.2	3	5	1	**0.0**	2	3
DPY-1	22	**0.2**	6	10	0.5	6	10	3	0.6	3	6
DPY-4	43	56.6	13	14	**2.7**	13	14	6	17.5	4	13
GSY-1	23	0.3	7	9	0.4	7	9	3	**0.2**	3	7
HDE	47	**1.0**	11	14	4.6	11	14	7	14.4	3	11
LAR	60	120.0	(gap=6.3%) 15	16	**5.9**	15	16	8	11.7	4	15
LDV	48	74.3	13	20	5.2	13	20	7	**2.8**	4	13
MAS	27	**0.5**	7	12	2.0	7	12	5	4.2	3	7
PS	17	0.4	5	6	0.5	5	6	2	**0.3**	3	5
PTH	19	**0.5**	5	10	0.8	5	10	2	**0.5**	3	5
RES	15	**0.2**	4	8	0.5	4	8	2	0.3	3	4
SGF	39	12.4	10	17	**2.5**	10	17	6	8.0	4	10
SO	44	3.9	10	15	**1.8**	10	15	7	7.2	3	10
THL-1	39	4.5	10	12	**1.7**	10	12	5	2.4	3	10
THL-2	28	3.0	7	10	**1.1**	7	10	3	4.0	3	7
V	12	**0.1**	4	7	0.3	4	7	2	**0.1**	3	4
VAI	27	**0.2**	6	10	0.6	6	10	4	0.6	3	6
VSG-REF	25	9.8	7	9	**0.6**	7	9	5	1.3	2	7
WPY-2	41	17.2	10	14	**3.3**	10	14	6	8.2	3	10
AMB-1	117	120.0	/	33	120.0	(gap=7.4%) 27	33	18	146.2	4	(gap=7.4%) 27
CLE	57	6.6	14	23	**5.9**	14	23	10	18.0	4	14
MOD	82	**2.3**	19	31	28.5	19	31	13	23.2	4	19
PPN	69	**2.2**	16	21	4.9	16	21	12	20.7	4	16
WPY-3	128	**1.5**	34	39	15.1	34	39	21	76.4	5	34

RCSPP: setting

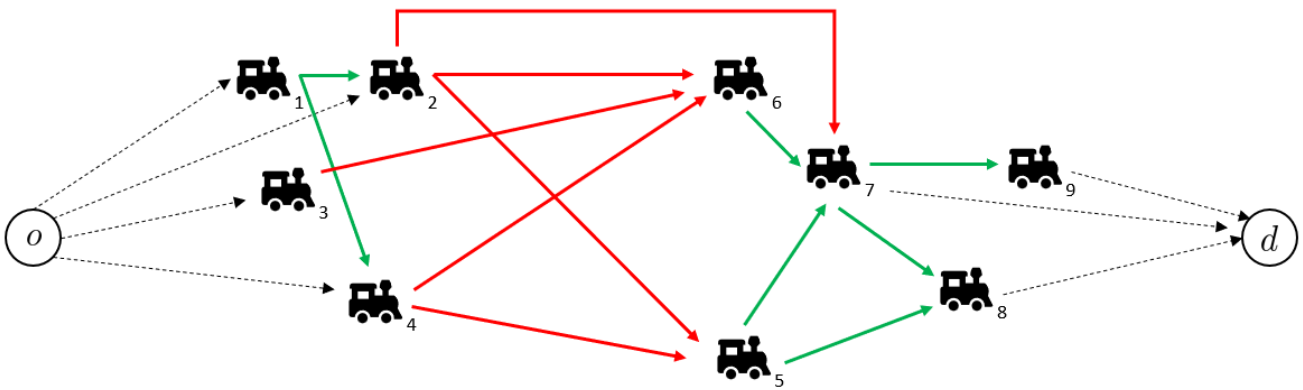
Input.

- Acyclic digraph with vertices o and d, partial ordered set of resources $(\mathcal{R}, \preccurlyeq)$
- Non-decreasing "extension function" on each arc $f_a : \mathcal{R} \to \mathcal{R}$ (describes evolution of a resource when arc a is added to a path $r_{P+a} := f_a(r_P)$) NON LINEAR
- Non-decreasing "cost & feasibility" function $c : \mathcal{R} \to \mathbb{R} \cup \{+\infty\}$ NON LINEAR

Output. Feasible o-d path P with minimum cost $c(r_P)$

e.g., shortest o-d path under time budget
→ *resource: (total cost, total time consumption)*
→ *extension function: (+arc cost, +arc time)*
→ *cost & feasibility function: total cost if total time consumption* $\leqslant$ *budget,* $+\infty$ *else*

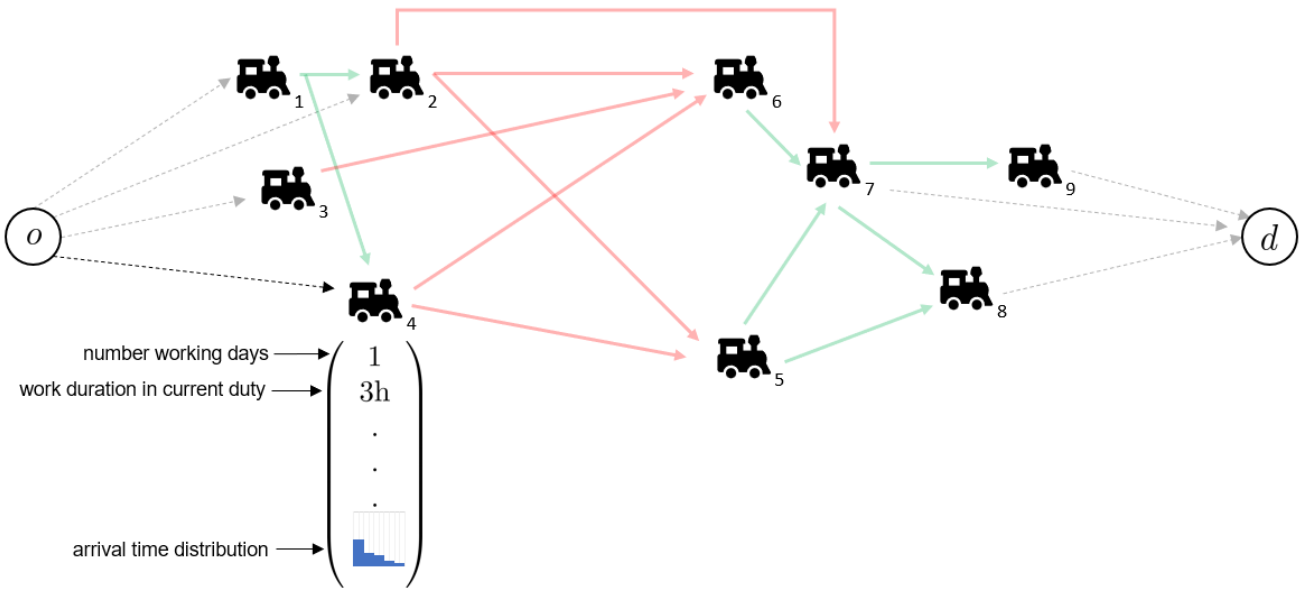
RCSPP: for the stochastic version



same duty

new duty

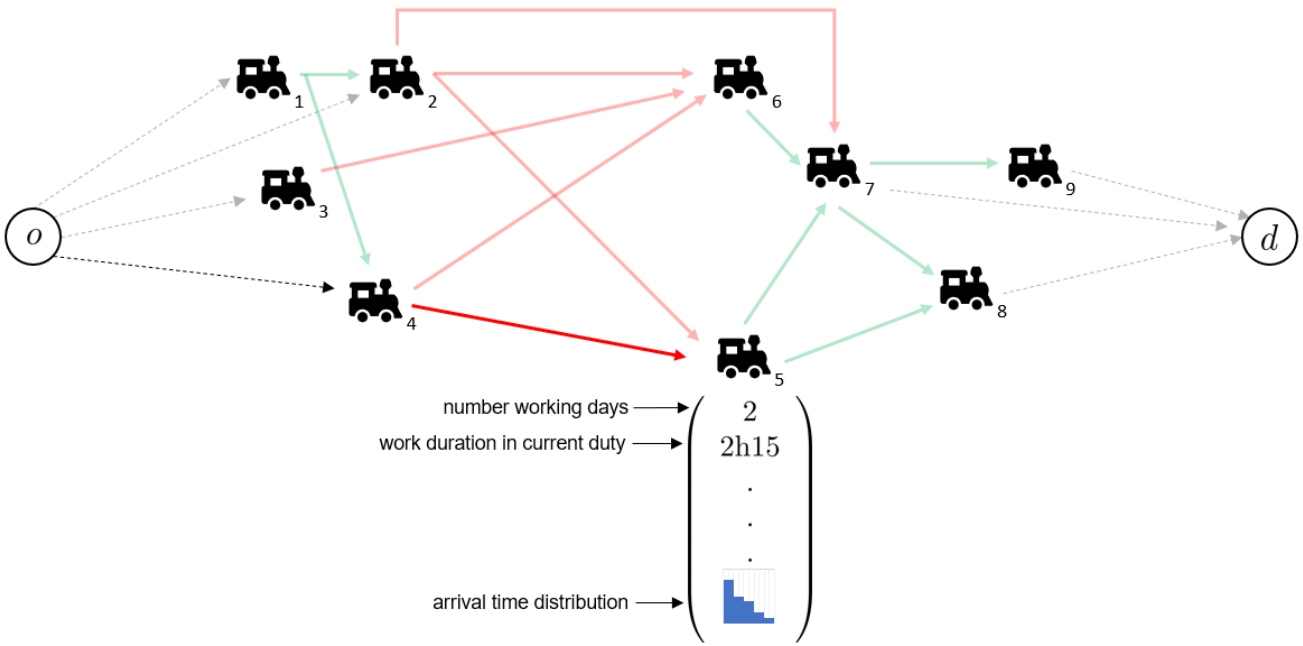
RCSPP: for the stochastic version



same duty

new duty

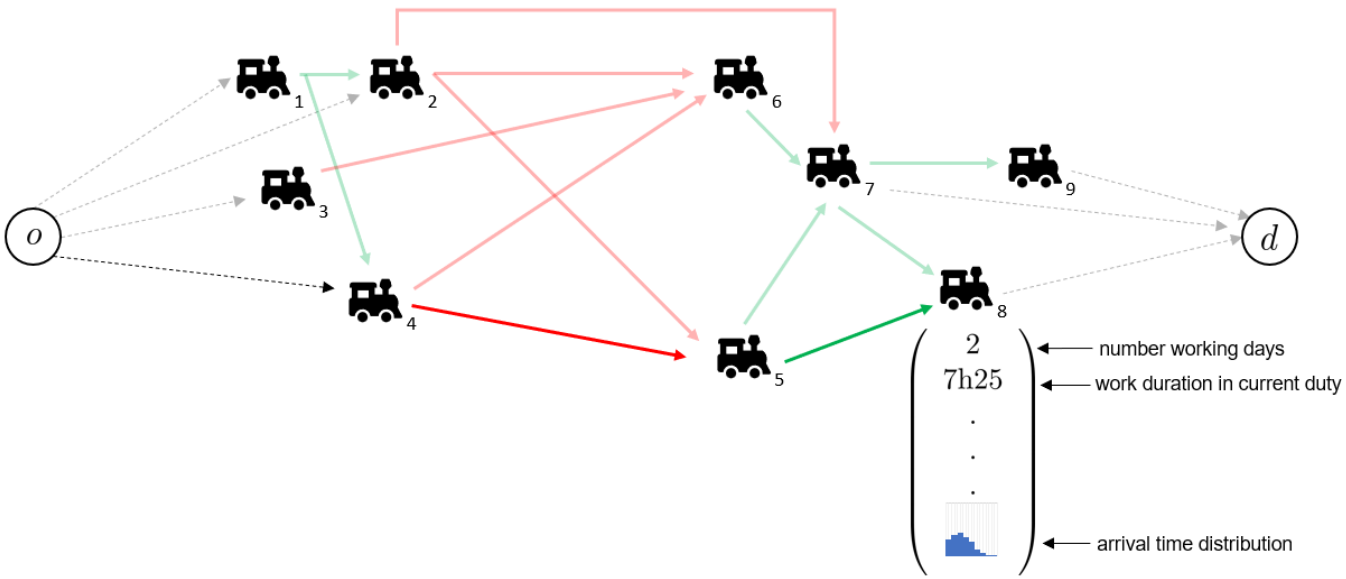
RCSPP: for the stochastic version



same duty

new duty

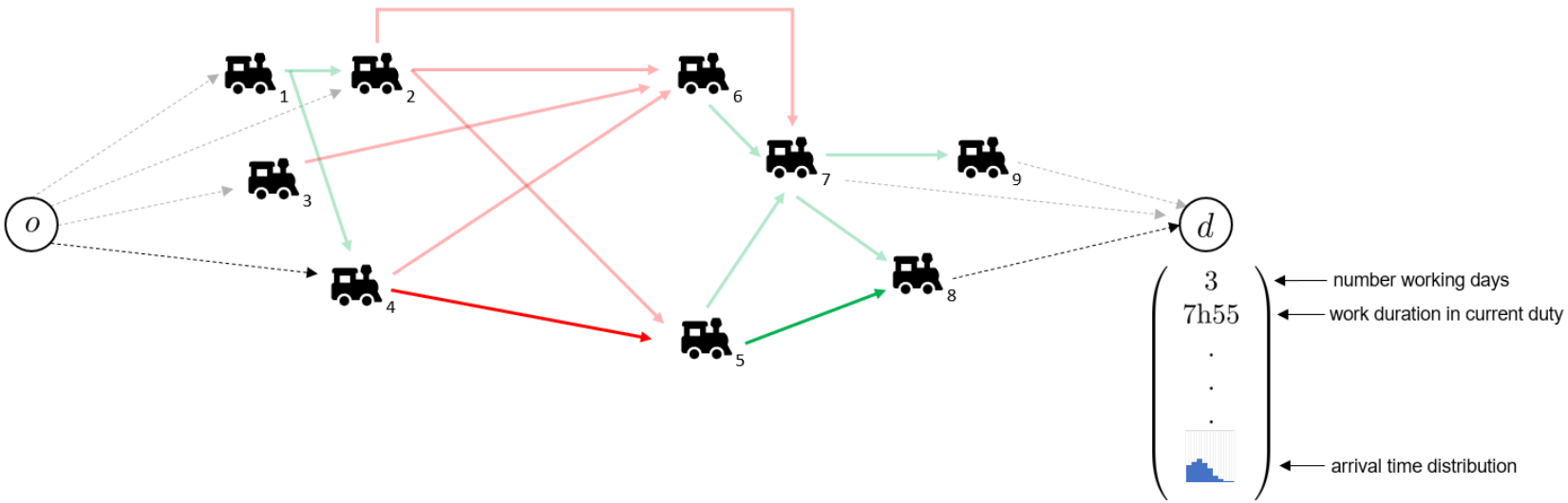
RCSPP: for the stochastic version



same duty

new duty

RCSPP: for the stochastic version



same duty

new duty

Improvements for the algorithm

- Early feasibility check
- Warm-starting the label correcting algorithm
- Set of bounds to discard paths
- Variable dominance from the master problem
- Better quality integer solution

Additional numerical results (Chapter 4)

	Number of variables	Number extended paths	Number discarded paths
116 trains	13,500	7,619,648	515,884,281
280 trains	16,821	7,749,733	519,288,691
926 trains	50,635	28,350,131	2,066,420,616
1,815 trains	83,133	50,359,197	4,062,062,782

Delay propagation: details

$$H_{t_i} = \begin{cases} \delta_{t_1} + \tau_{t_1} + P_{t_1} & \text{if } i = 1, \\ \max\left(H_{t_{i-1}} + \Delta_{t_{i-1},t_i}, \delta_{t_i}\right) + \tau_{t_i} + P_{t_i} & \text{if } i \geqslant 2, \end{cases}$$

- Δ_{t_{i-1},t_i}: minimal time to operate consecutively the trains t_{i-1} and t_i taking into account possible relocations and short breaks
- δ_t: nominal departure time of train t
- τ_t: nominal duration of train t

Fragility: details

$$\sigma(h, b) = \sum_{i=1}^{q} \mathbb{1}\left(h_{t_{i-\mathbf{1}}, k_{i-\mathbf{1}}} > s_i \text{ or } \max_{j=2,\ldots,k_i} \left(h_{t_{i,j-\mathbf{1}}} + \Delta_{t_{i,j-\mathbf{1}}, t_{i,j}} \right) > s'_i \right),$$

- duty sequence of $b = d_1, d_2, \ldots, d_q$
- train sequence of each duty $d_i = t_{i,1}, t_{i,2}, \ldots, t_{i,k_i}$
- s_i: instant beyond which arrival of last train of duty d_{i-1} makes the duty d_i disrupted
- s'_i: instant beyond which departure of a train makes the duty d_i disrupted

Variant of the stochastic crew management problem

$$
\begin{aligned}
\text{minimize} \quad & \sum_{i \in I} \sum_{b \in B_i} \mathbb{E}[\sigma(b, H)] x_{b,i} \\
\text{s.t.} \quad & \sum_{i \in I} \sum_{b \in B_i : t \in b} x_{b,i} \geqslant 1 && \forall t \in T \\
& \sum_{i \in I} \sum_{b \in B_i} w_b x_{b,i} \leqslant W \\
& x_{b,i} \in \{0, 1\} && \forall i \in I, \forall b \in B_i
\end{aligned}
$$

	Two-stage approach			Integrated approach: stochastic case				
	Time	Nb working days	Fragility	Threshold W	Time	Nb working days	LB	Fragility UB
116 trains	1 min	50	0.970	50	26 min	50	0.0	0.0
280 trains	3 min	228	2.450	228	4 min	228	0.0	0.0
926 trains	26 min	832	8.538	832	2 h 37 min	832	0.0	0.3
1,815 trains	64 min	1,633	14.900	1,633	1 h 45 min	1,633	0.0	0.0
1,815 trains bis	64 min	1,633	18.410	1,633	2 h 24 min	1,633	0.0	0.0

Line planning: problem statement

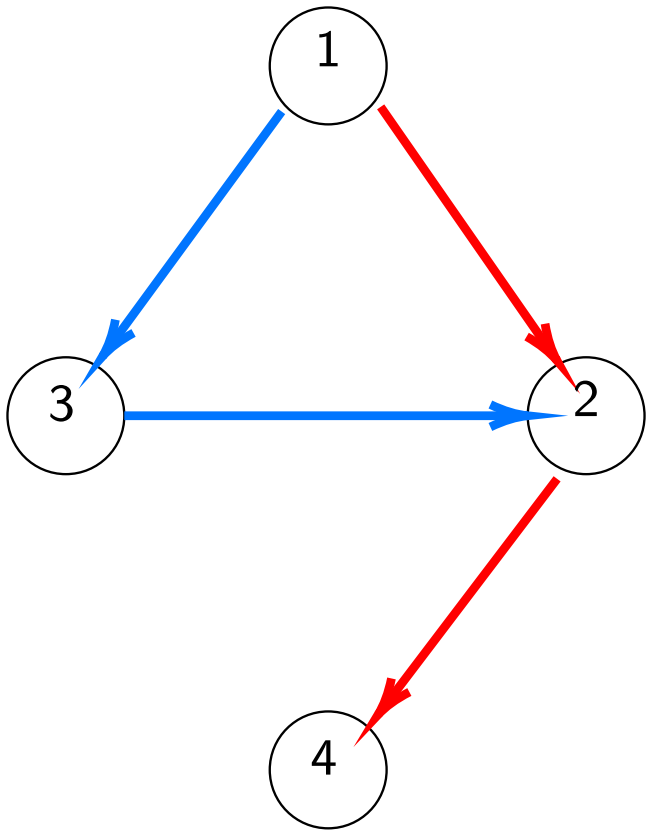
Input.

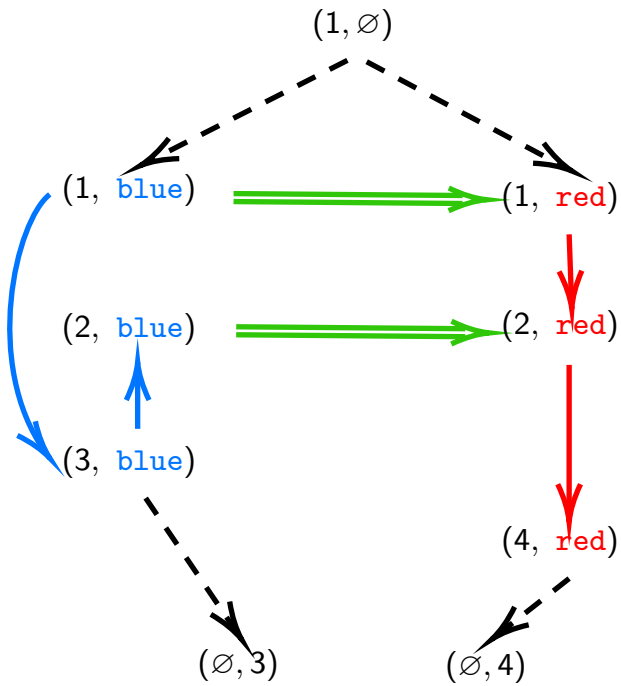
- public transportation network
- demands: number of users between each $o - d$ (in planning horizon)
- line pool, each line serving a sequence of stations

Output. Choose the lines to operate and their frequency (in planning horizon) such that:

- all demands can be met
- the budget B for line opening is not exceeded
- the maximum vehicle frequency over each arc is not exceeded
- the overall users inconvenience is minimal

Change-and-go network





Arc-based formulation

$$\text{minimize} \sum_{(o,d)\in\mathcal{D}} \sum_{a\in\mathcal{A}} c_a x^a_{o,d}$$

$$\text{s.t.} \sum_{(o,d)\in\mathcal{D}} x^a_{o,d} \leqslant N_\ell f_\ell \qquad \forall \ell \in \mathcal{L}, a \in \mathcal{A}^\ell_{\text{go}}$$

$$\sum_{\ell\in\mathcal{L}} C_\ell f_\ell \leqslant B$$

$$\sum_{\ell\in\mathcal{L}:\ a\in A(\ell)} f_\ell \leqslant f^{\max}_a \qquad \forall a \in A$$

$$\sum_{a\in\delta^+((o,\varnothing))} x^a_{o,d} = w_{o,d} \qquad \forall (o,d) \in \mathcal{D}$$

$$\sum_{a\in\delta^-((\varnothing,d))} -x^a_{o,d} = -w_{o,d} \qquad \forall (o,d) \in \mathcal{D}$$

$$\sum_{a\in\delta^+(v)} x^a_{o,d} - \sum_{a\in\delta^-(v)} x^a_{o,d} = 0 \quad \forall (o,d) \in \mathcal{D}, v \in \mathcal{V}\backslash\{(o,\varnothing),(\varnothing,d)\}$$

$$x^a_{o,d} \in \mathbb{Z}_{\geqslant 0} \qquad \forall (o,d) \in \mathcal{D}, a \in \mathcal{A}$$

$$f_\ell \in \mathbb{Z}_{\geqslant 0} \qquad \forall \ell \in \mathcal{L}$$

Path-based formulation: Dantzig–Wolfe formulation

$$\text{minimize} \sum_{(o,d)\in\mathcal{D}} \sum_{p\in P_{o,d}} c_p w_{o,d} \lambda^p_{o,d}$$

$$\text{s.t.} \sum_{(o,d)\in\mathcal{D}} \sum_{p\in P_{o,d}\,:\, a\in p} w_{o,d}\lambda^p_{o,d} \leqslant N_\ell f_\ell \qquad \forall \ell \in \mathcal{L}, a \in \mathcal{A}^\ell_{\mathbf{go}}$$

$$\sum_{p\in P_{o,d}} \lambda^p_{o,d} = 1 \qquad \forall (o,d) \in \mathcal{D}$$

$$\sum_{\ell\in\mathcal{L}} C_\ell f_\ell \leqslant B$$

$$\sum_{\ell\in\mathcal{L}\,:\, a\in A(\ell)} f_\ell \leqslant f^{\max}_a \qquad \forall a \in A$$

$$\sum_{p\in P_{o,d}\,:\, a\in p} w_{o,d}\lambda^p_{o,d} \in \mathbb{Z}_{\geqslant 0} \qquad \forall (o,d) \in \mathcal{D}, a \in \mathcal{A}$$

$$f_\ell \in \mathbb{Z}_{\geqslant 0} \qquad \forall \ell \in \mathcal{L}$$

$P_{o,d}$: set of elementary paths from $(o, \varnothing)$ to $(\varnothing, d)$

Methodology

Column generation with as many pricing subproblems as *o*-*d* pairs

Pricing subproblem: Shortest path problem solved using Dijkstra's algorithm

Issue: Retrieving a feasible integer solution

Branch-and-price: branching, inequalities and diving

Branching:

- Frequency variables depending on the middle of its domain

Valid inequalities:

- Knapsack inequalities for lines with almost same cost
- Extended cover inequalities for integer knapsack
- Aggregated inequalities for frequencies from origin node

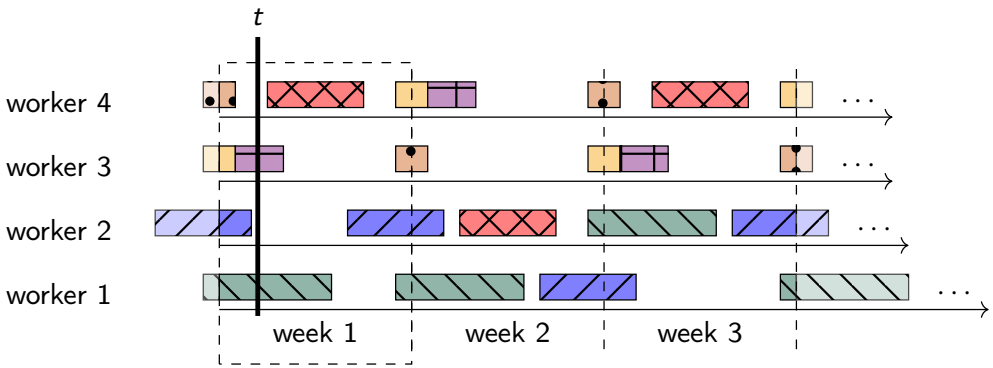
Diving:

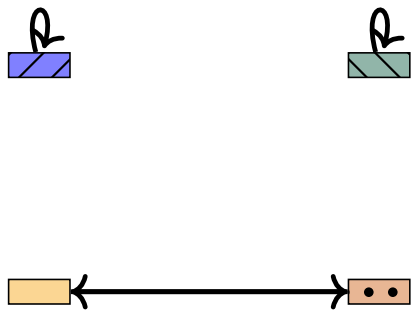
- if there exists a line for which rounding down the frequency does not violate the vehicle capacity constraint in the current fractional solution, then we round its frequency down
- if there exists a line for which rounding up the frequency does not violate the budget constraint in the current fractional solution (taking into account the cost reduction of rounding down the line frequencies from the first item), then we round its frequency up

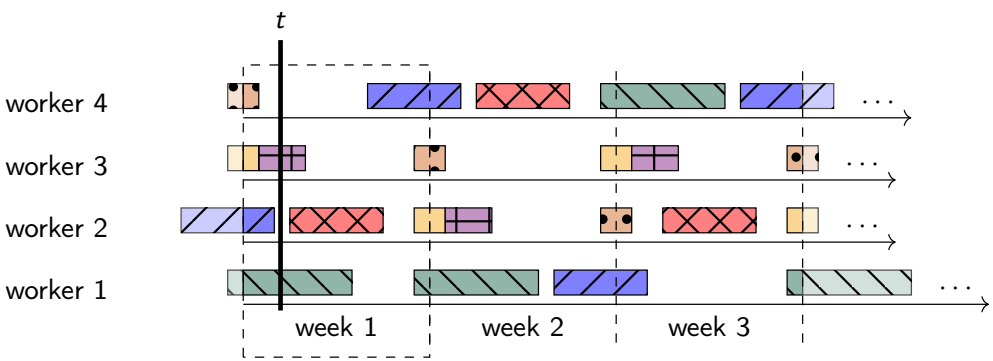
Numerical experiments

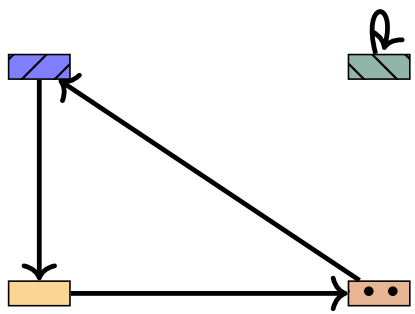
	Arc-based: ILP			Path-based: Branch-and-price-and-cut						
	Time (s)	Best integer	Gap (%)	Root node			Total time (s)	Best integer	Gap (%)	Number nodes
				Time (s)	Value	ILP value				
small	0.1	1,048.0	0.0	0.02	1,048.0	1,048.0	0.2	1,048.0	0.0	1
toy	0.1	44,217.0	0.0	0.4	44,217.0	44,217.0	0.4	44,217.0	0.0	1
mandl	3.0	327,430.0	0.0	2.2	326,830.0	/	3,600.0	327,430.0	0.1	865
grid	1.4	76,834.0	0.0	1.8	76,834.0	76,834.0	1.8	76,834.0	0.0	1
athens	103.6	17,569,819.0	0.0	297.8	17,569,819.0	/	3,600.0	/	$+\infty$	74
ring	memory overload			285.5	102,438.8	/	3,600.0	/	$+\infty$	8
bahn-01	memory overload			1,202.4	13,442,535.0	/	3,600.0	/	$+\infty$	4

Merge operations

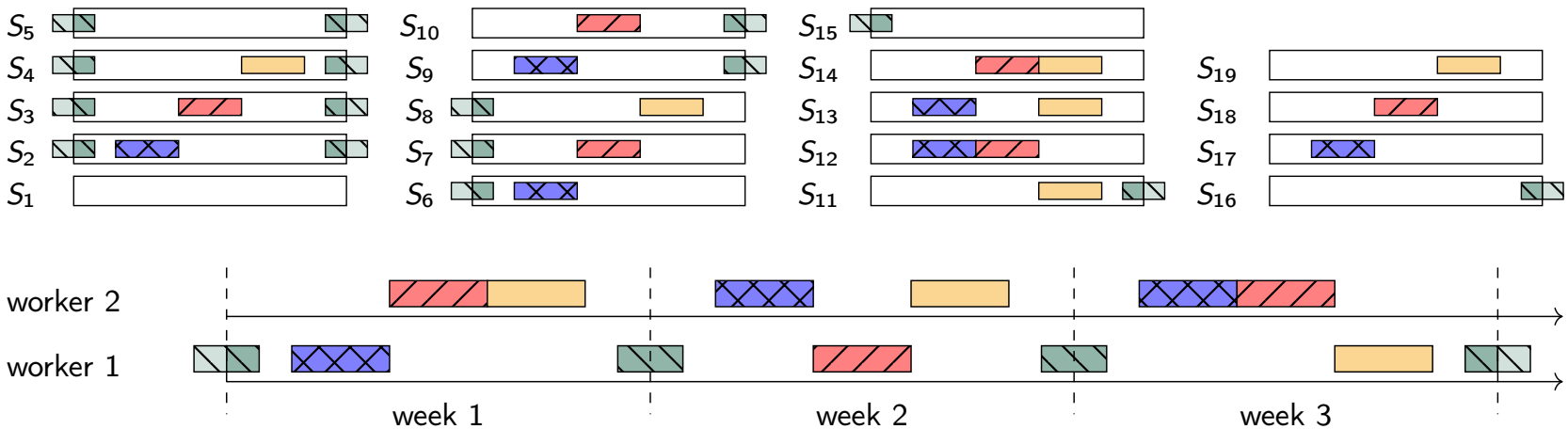








Example: why the extra condition is needed



Proof of Proposition 8.3.1

Let G be a group and S be a generating set of G. The Cayley graph $\Gamma(G, S)$ is an edge-colored directed graph constructed as follows:

- Each element g of G is assigned a vertex: the vertex set of $\Gamma(G, S)$ is identified with G.
- Each element s of S is assigned a color c_s.
- For every $g \in G$ and $s \in S$, there is a directed edge of color c_s from the vertex corresponding to $g \in G$ to the one corresponding to gs.

For each vertex $i \in V$, define $\widetilde{p}_i \colon \Gamma(\langle\Sigma\rangle, \Sigma) \to D$ as follows:

- for a vertex g of $\Gamma(\langle\Sigma\rangle, \Sigma)$, set $\widetilde{p}_i(g) := g(i) \in V$,
- for an arc $\tilde{a}$ of $\Gamma(\langle\Sigma\rangle, \Sigma)$ going from g to $\sigma_c \circ g$ with color c, set $\widetilde{p}_i(\tilde{a})$ as the arc in A going from $g(i)$ to $(\sigma_c \circ g)(i)$ with color c.

Lemma 8.5.1, p.133

We have $|\widetilde{p}_i^{-1}(a)| = |\langle\Sigma\rangle|/|V|$ for every $i \in V$ and $a \in A$.

Eulerian cycle on the Cayley graph $\Gamma(\langle\Sigma\rangle, \Sigma)$ works

Particular case: problem of pebbles on a Cayley graph

Lemma T. Tao

A colored directed graph is a Cayley graph if and only if the three following properties hold:

- (Regularity) For each color c, every vertex i has a single arc with head i and color c, and a single arc with tail i and color c.
- (Connectedness) The graph is connected.
- (Homogeneity) For every pair of vertices i, j, there is a unique colored graph isomorphism that maps i to j.

Proposition 9.1.2, p.138

If D is a Cayley graph, then there exists a periodic balanced sequence of colors with period $|A|$.

Proof. D is isomorphic to $\Gamma(\langle\Sigma\rangle, \Sigma)$.

Relaxed sequence of colors problem

Periodic balanced assignment (PBA)
Input. Tasks $[n]$, number q of workers, set $\mathcal{S}$ of non-overlapping schedules
Ouput. A periodic balanced feasible assignment with minimal period

Periodic relaxed balanced sequence of colors (PRBC)
Input. A regular colored Eulerian digraph $D = (V, A)$ such as no two colors encode the same permutation
Ouput. A finite sequence of colors of minimal length that makes each pebble follow a closed walk on D, and such that, within each color, each pebble visits each arc the same number of times

Proposition 9.2.1, p.139

Given an instance $D = (V, A)$ of the (PRBC) problem with set of colors C, we can build an instance $\mathcal{I}(D)$ of the (PBA) problem with $q = |V|$, $|\overline{\mathcal{F}}_{\mathcal{I}(D)}| = |C|$, and same optimal value in polynomial time.

ILP formulation for a (relaxed) balanced sequence colors

$$\text{minimize} \sum_{c\in C}\sum_{k\in[N]} y_{c,k}$$

$$\text{s.t.} \sum_{k\in[N]} x_{p_1,a,k} = \sum_{k\in[N]} x_{p_2,a,k} \qquad \forall (p_1,p_2)\in P^2, a\in A$$

$$\sum_{k\in[N]} x_{p,a_1,k} = \sum_{k\in[N]} x_{p,a_2,k} \qquad \forall c\in C, p\in P, (a_1,a_2)\in A_c^2$$

$$x_{p,a,k} \leqslant \sum_{c\in C} y_{c,k+1} \qquad \forall k\in[N-1], p\in P, a\in A,$$

s.t. head(a)$\neq$initial vertex for p

$$x_{p,a,N} = 0 \qquad \forall p\in P, a\in A$$

s.t. head(a)$\neq$initial vertex for p

$$\sum_{c\in C} y_{c,k+1} \leqslant \sum_{c\in C} y_{c,k} \qquad \forall k\in[N-1]$$

$$\sum_{c\in C} y_{c,1} = 1$$

$$\sum_{c\in C} y_{c,k} \geqslant \sum_{a\in A} x_{p,a,k} \qquad \forall k\in[N], p\in P$$

$$y_{c,1} = x_{p,a,1} \qquad \forall c\in C, p\in P,$$

$a =$ arc of color c s.t. tail(a)=initial vertex for p

$$y_{c,k+1} + x_{p,a,k} - 1 \leqslant x_{p,a',k+1} \qquad \forall c\in C, k\in[N-1], p\in P, a\in A$$

$a' =$ arc of color c s.t. tail(a')=head(a)

$$x_{p,a,k}\in\{0,1\} \qquad \forall p\in P, a\in A, k\in[N]$$

$$y_{c,k}\in\{0,1\} \qquad \forall c\in C, k\in[N]\,.$$

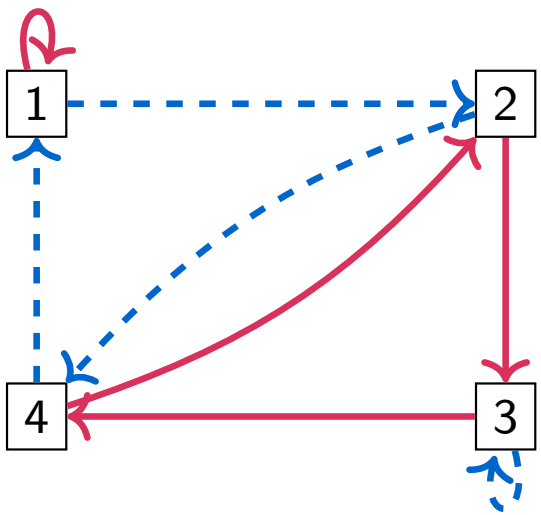
ILP formulation for a (relaxed) balanced sequence colors

Lemma 9.3.2, p.144

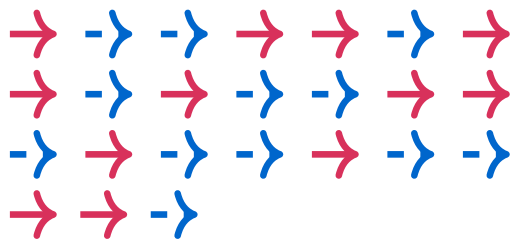
The following constraints are valid inequalities for the ILP:

$$\sum_{c \in C} (y_{c,k} - y_{c,k+1}) = 0 \quad \forall k \in [N-1] \colon k \equiv 0 \pmod{|P|}.$$

Examples reaching the balanced sequence of colors bound

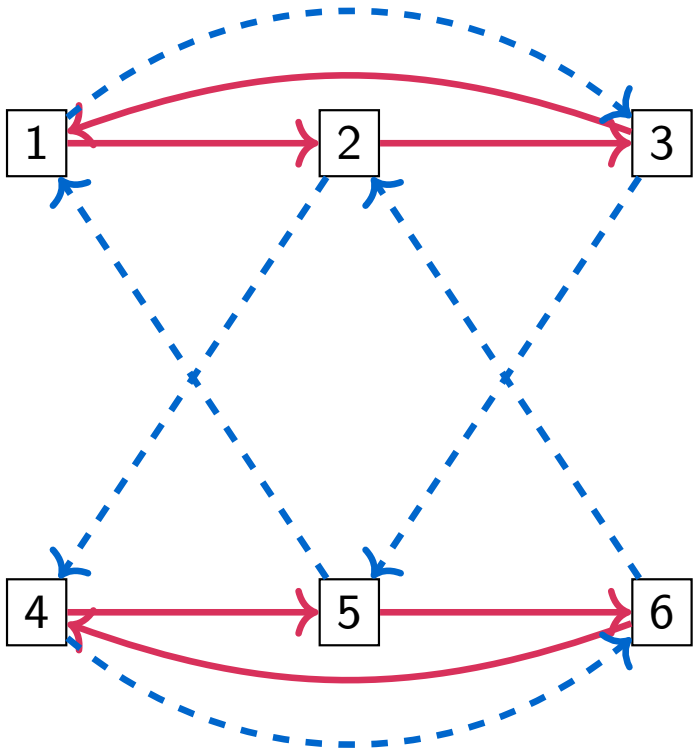


Shortest relaxed balanced sequence of colors:

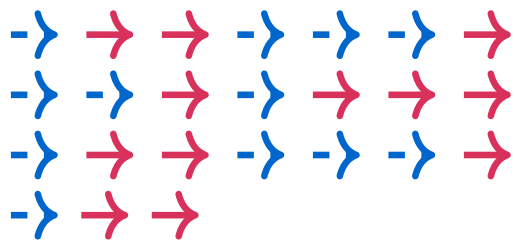


Bound:
$|A| \times |\langle \Sigma \rangle| / |V| = 24$

Examples reaching the balanced sequence of colors bound



Shortest relaxed balanced sequence of colors:



Bound: $|A| \times |\langle \Sigma \rangle| / |V| = 24$

A problem of minimal frequencies over skills acquired by tasks

A feasible assignment achieves the frequencies $\nu_k \in \mathbb{R}_{\geqslant 0}$ for $k \in K$ if for every $k \in K$ and $j \in [q]$:

$$\lim_{t \to +\infty} \frac{1}{t} \sum_{r=1}^{t} \sum_{i=1}^{n} \alpha_k^{(i)} \mathbb{1}(f(i,r) = j) \geqslant \nu_k \,.$$

Feasible assignment in the extended version: complexity

Lemma

Deciding whether there exists a feasible assignment in the extended version is NP-hard.

Proof. Exact cover problem
Input. Collection P of subsets of a set X
Output. A subcollection $P^\star$ of P such that: the subsets in $P^\star$ are pairwise disjoint and cover X